Author's accepted manuscript. G. Chen, J. Luo, S. Yao, H. Yuan, Y. Wang, T. Yu, H. Wang and Z. Lin, Contact sensing methodology based on propagation of interaction-induced deformation in elastic beams, The International Journal of Robotics Research (2026). Copyright © 2026 The Author(s). DOI: 10.1177/02783649261419836

The International Journal of Robotics Research
XX(X):1–18
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DOI: 10.1177/ToBeAssigned
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Contact Sensing Methodology Based on Propagation of Interaction-Induced Deformation in Elastic Beams

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Abstract

Owing to the capability of environment perception, contact sensors are increasingly used in robotic systems to enhance their ability to interact with the environment. This paper proposes a simple contact sensing methodology based on the continuum mechanics of largely deformed beams. Slender elastic beams embedded with flexible strain gauges are employed as the sensing medium for contact perception. An explicit model characterizing the beam's force-deflection behavior is established to map the local bending of strain gauges to the contact-induced global deformation. Thus, the exact status of contact positions and forces can be determined efficiently. A variety of experiments are conducted to validate the effectiveness of the developed contact sensor. In addition to contact perception, the elastic beam endows robotic systems with structural compliance for interactive manipulation. Hence, applications for object size identification, active control of contact force, and force-based collaboration are provided to exhibit the potential of the proposed contact sensing methodology in mobile robots.

Keywords

Contact sensing, continuum mechanics, environment perception, object recognition, interactive manipulation.

1 Introduction

Detecting external contacts equips robotic systems with the ability to perceive the surrounding environment through physical interaction (Sundaram et al. 2019; Johansson and Flanagan 2009; Chun et al. 2021). With the exact contact information acquired from the sensors, robots can detect mechanical stimuli and quantify the interaction in a more accurate and intuitive way (Dahiya et al. 2009; Schmitz et al. 2011; Luo et al. 2017). Benefiting from this capability, robots that integrate contact-based sensors are able to broaden their potentials in various scenarios of applications, such as object recognition (Xiao et al. 2022; Bauza et al. 2023), dexterous manipulation (Lloyd and Lepora 2021; Chitta et al. 2011), and human-machine interactions (Leboutet et al. 2019; Ku et al. 2024). However, it still remains a great challenge for different contact sensors to be used in practical applications, due to existing issues, such as the complexity and difficulty in manufacturing, the reliability and durability for usage, and as well, the sensitivity and precision of contact estimation (Kappassov et al. 2015; Li et al. 2020). To tackle the above problems, great efforts have been made in the past decades, in promoting the techniques to detect contacts in robotics.

For robots consisting of rigid links, a typical strategy for sensing external contact is to attach force sensor to the end-effector, where the contact force and torque can be directly measured (Zhang et al. 2020). This approach is known as force sensing methodology (Siciliano et al. 2008). As well, torque and/or force sensors can be integrated into the active joints of a robot, such as most of the collaborative robots, in which the capability of contact sensing can be expanded from the end-effector to the whole robotic arm (Kim et al. 2018; Sun et al. 2015). As traditional force/torque sensors can only implement measurements for concentrated loads, shells equipped with flexible circuits and vision trackers, are constructed as the contact sensor for rigid robots (Cirillo et al. 2015; Kim et al. 2021; Duong and Ho 2020; Laliberté and Gosselin 2021), to enable them with the ability to perceive external loads exerted in various locations. Force sensors have also been integrated into robotic systems with flexible links to endow the capability of contact sensing (Pandya et al. 2016). In these cases, force sensors are usually equipped on the actuators or on the end-effector (Kübler et al. 2005; Valdastri et al. 2006). To detect the contact occurring in locations other than the end-effector, optoelectronic sensors are embedded into the inner segments of flexible robots (Noh et al. 2016, 2019) to achieve wide-range contact perception. However, the direct use of force sensors on robotic systems can only detect contact occurring in limited locations, which restricts the capability of perception.

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An alternative approach widely employed for contact detection is tactile sensing methodology, where spatially resolved transduction of mechanical quantities—such as normal and shear stress, skin deflection, or thermal gradients—across an extended compliant surface are studied. To achieve tactile sensing, electronic skins (e-skins) are commonly used to perceive and quantify external mechanical stimuli from the electronic signals of an array of sensors based on different principles (Pyo et al. 2021; Chortos et al. 2016; Wang et al. 2015). Pressure sensors are usually adopted in e-skins to measure the contact location and extent in the normal direction. Su et al. (2021) developed a stretchable and sensitive capacitive pressure sensor to realize sensation of interactions on humans or soft robotic skins. Using flexible piezo-resistive pressure sensors, Gong et al. (2014) developed a sensitive e-skin capable of vibration detection. In addition, tactile sensors that are able to measure both the normal and shearing forces, are proposed by Yan and his colleagues (2021) for robust grasping. By compositing the temperature, acceleration, and proximity sensors, Mittendorfer and Cheng (2018; 2011) developed a multifunctional e-skin for humanoid robots to realize recognition of various objects. In general, tactile sensing of e-skins relies on an array of independent sensors whose signals are measured in an individual manner. As a consequence, the complexity in both structure and signal acquisition increases significantly as large-scale e-skins of high-resolution are required.

Distinct from force sensing and tactile sensing methodologies, another paradigm in external contact detection focuses on estimating both contact locations and force magnitudes, which is defined as contact sensing. To achieve agile perception of unstructured environments, the concept of whisker-type sensors are proposed for contact sensing on a variety of robots, by means of mapping the contact-induced strain, bending moment or deformation to the external mechanical stimuli (Diamond et al. 2008; Sayegh et al. 2022; Solomon and Hartmann 2006, 2010; Bebek and Cavusoglu 2008; Patanè et al. 2012). Via biomimetic whisker sensors, contact sensing can be realized by touching the environment in either a passive or an active way. An and his colleagues (2021) developed a bendable whisker mechanoreceptor for mobile robots as contact sensors, which are used for obstacle recognition, and object surface scanning and reconstruction, in a passive way. By adding an extra actuator, Yu et al. (2022) developed an active whisker sensor for distance, shape, size, and orientation sensing. Thus, high-efficient contact sensing could be achieved for mobile robots by sweeping the hairy whiskers actively to interact with the environment. Although the robotic whisker sensors are usually simple in structure, less information could be acquired from each single whisker, as only a few receptors are placed at the base for most whisker sensors. While some sensor designs have demonstrated the capability to localize contact points in the intermediate region of whisker structures (Sofla et al. 2023, 2024), these implementations typically require cumbersome force/torque measurement instrumentation and extensive sampling apparatus, thereby compromising the overall system's compactness and mobility.

Otherwise, contact sensing methodology based on elastic mechanics of flexible beams or rods is able to estimate the exact deformed status in contact with the environment. Under the constant-curvature assumption, Simaan et al. (2008; 2010) proposed a simple model to predict the exact external load exerted at the tip of a flexible rod. Based on the Correrat rod theory, Rucker and his colleagues (2017) developed a model-based approach for the estimation of geometric constraints to the distal platform, according to the actuation forces measured at the base. In such cases, contact-induced force/torque can be used to estimate the interaction of the elastic structure with the environment, which are classified as intrinsic sensing (Bicchi 1990). Through a vision system, Hooshiar et al. (2021) developed an algorithm to estimate the contact force by capturing the tip pose. Similarly, electromagnetic sensors (Back et al. 2016; Yuan et al. 2017; Back et al. 2018) are also used to track the tip pose for estimating the contact forces. As well, the information of local deformation, measured via vision systems (Back et al. 2015; Razban et al. 2018) or fiber Bragg grating sensors (Qiao et al. 2021), can be extracted to identify the exact contact at the tip or intermediate sections. Although accurate results can be obtained using these model-based approaches, the introduction of external measurement devices severely affects the compactness of the contact sensors and limits their application scenarios.

In this paper, a simple yet reliable contact sensing method is proposed based on the continuum mechanics of slender elastic beams. Then, an explicit model that characterizes the force-deflection behavior of the constrained elastic beams is established to calculate the contact-induced deformations along the slender beams efficiently. In the developed contact sensors, industrial-grade thin-film strain gauges are used to make them compact in structure, stable in sensing, and easy for deployment. Hence, only a small number of strain gauges are required to endow the elastic beam with the capability of multi-point contact sensing along the whole elastic structure. Owing to utilization of highly-elastic material and industrial strain gauges, the developed sensors are of high-frequency response and low-hysteresis characteristics. Experiments are conducted to validate the effectiveness of the contact sensing methodology. Meanwhile, the slender elastic beam not only serves as a contact sensor, but also acts as a compliant unit to buffer external impact. Thus, it endows robotic systems with the ability to implement force-based interactive manipulation with the environment and other robots. Demonstrations are provided to exhibit enormous potential in robotic systems for force-based perception and manipulation.

This paper is organized as follows: Section 2 presents the measuring method and modeling framework of the proposed contact sensing methodology. Then in Sec. 3, a kinetostatic model is established in closed-form based on the continuum mechanics of the largely-deformed elastic beams. Section 4 provides the modeling of contact sensing in both scenarios of small-deformation contact and large-scale interaction. To verify the feasibility and efficiency of the developed contact sensor, a variety of experiments are conducted on a prototype developed in Sec. 5. A couple of examples are provided in Sec. 6, to demonstrate the potential of the proposed contact sensors applied in mobile robots for environment perception and interactive collaboration. Finally, conclusions are drawn in Sec. 7.

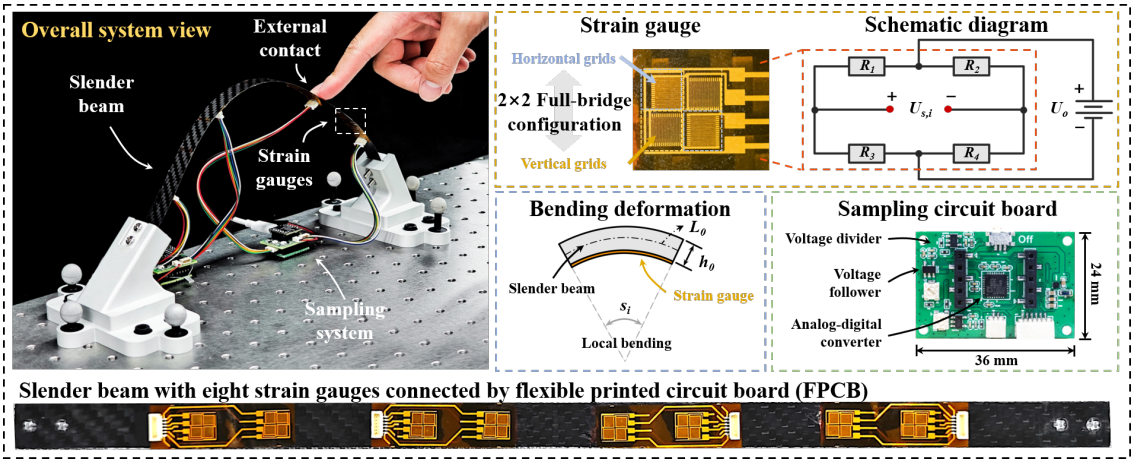

Figure 1. Measuring system particularly developed for the proposed contact sensing methodology.

2 Sensing Method and Modeling Framework

This section presents the kinetostatics modeling and analysis of the elastic beams that are utilized as the contact sensors. A discretization-based approach (Chen et al. 2015, 2018) is introduced to establish the kinetostatic model of the slender elastic beam, and characterize the force-deflection behavior. Then, the overall contact stiffness can be derived concisely in an analytical way, which serves as the fundamental for the contact sensing methodology proposed in this paper.

2.1 Measuring System of the Beam's Local Deformations

In the proposed contact sensing method, special strain gauges are particularly fabricated to measure the local deformations of the slender elastic beam. As shown in Fig. 1, a full-bridge configuration, the 2×2-grid form, is adopted for the strain gauges to reduce the influence of measuring noise. Owing to such a design, the sampling voltage are linearly related to its bending, and as well, is robust to temperature drift.

As illustrated in Fig. 1, eight strain gauges are deployed on the elastic beam at discrete locations. To keep the measuring system compact, a thin flexible printed circuit board (FPCB) is used to connect the strain gauges to the sampling circuit. Thus, the extra constraint induced by the measuring system is neglectable to the beam's kinetostatics. Further, all devices, including the low-pass filter, amplifier, and A/D converter, are integrated on the same board to reduce space. As a result, the bending angles associated with the strain gauges can be obtained according to the change of voltages measured by the sampling circuit. And this provides the response datum for the proposed contact sensing methodology. Moreover, the relationship between the local bending and the measured voltage of the strain gauges can be represented in a linear form (Luo et al. 2022) as

$$s_k = \frac{8 l_0}{K h_0 U_0} U_{s,k} = \lambda_k U_{s,k} \tag{1}$$

where U_0 and $U_{s,k}$ denote the supply and measured voltages of the k^{th} strain gauge. Here, $K = (1+2\mu) + (1-2\mu)C$ is the sensitivity ratio, with μ and C being the Poisson's ratio and Bridgman's constant of the wire material, respectively.

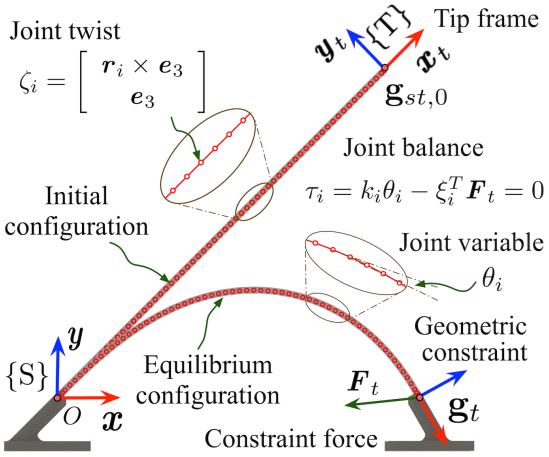

Figure 2. Elastostatics modeling of the beam's elastic component.

While h_0 and l_0 represent the thickness of the beam and the effective length of the gauges, respectively.

2.2 Modeling and Analysis of Large-Deformation Problems

Using the proposed method, the slender elastic beam will be discretized to a number of small segments, whose structural compliance is characterized by passive elastic joints. Thus, the whole beam can then be approximated by an articulated linkage consisting of rigid bodies and elastic joints. Hence, the beam's large deformation problems can be modeled as the kinetostatics of the equivalent linkage and solved readily using the conventional algorithms. In this work, the product-of-exponential (POE) formulation (Murray et al. 1994) is adopted to establish the kinematic and static models of the equivalent linkage.

As drawn in Fig. 2, the slender beam is geometrically constrained to the fixed base at both the proximal and distal ends. Using the POE formula, the geometric constraint to the tip frame of its equivalent linkage (Chen et al. 2014) can be represented as

$$\mathbf{g}_{st}(\boldsymbol{\theta}) = \exp(\hat{\zeta}_1 \theta_1) \cdots \exp(\hat{\zeta}_n \theta_n)\, \mathbf{g}_{st,0} = \mathbf{g}_t \tag{2}$$

where $\mathbf{g}_{st} \in SE(3)$ represents the relative pose of the tip frame $\{\mathrm{T}\}$ with respect to the spatial one $\{\mathrm{S}\}$. $\hat{\boldsymbol{\zeta}}_i \in se(3), i = 1, \cdots, n$ are the twists, in the 4×4 standard representation, of the elastic joints. While $\boldsymbol{\theta} = [\theta_1, \cdots, \theta_n] \in \mathbb{R}^{n\times1}$ corresponds to the joint displacements. Here, $\mathbf{g}_{st,0} \in SE(3)$ relates to the initial pose of $\{\mathrm{T}\}$ at $\boldsymbol{\theta} = \boldsymbol{0}$ and $\mathbf{g}_t$ denotes its target one.

Since the joints in the equivalent linkage are passive but elastic, the static equilibrium should be taken into account in the kinetostatics analysis. It implies that the restoring torques generated in these elastic joints are exactly balanced by those transmitted from the constraint force reacted at the distal end. As a result, the equilibrium condition in the joint space of the linkage can be represented in a compact form as

$$\boldsymbol{\tau} = \mathbf{K}_{\boldsymbol{\theta}}\,\boldsymbol{\theta} - \mathbf{J}_t^T\,\boldsymbol{F}_t = \boldsymbol{0} \tag{3}$$

where $\mathbf{K}_{\boldsymbol{\theta}} = \mathrm{diag}(k_1, \cdots, k_n) \in \mathbb{R}^{n\times n}$ is the diagonal stiffness matrix of the elastic joints. $\boldsymbol{F}_t \in \mathbb{R}^{6\times1}$ denotes the constraint wrench exerted at $\{\mathrm{T}\}$. And $\mathbf{J}_t \in \mathbb{R}^{6\times n}$ represents the Jacobian matrix relating the joint space to the tip frame.

Using POE formulation, the Jacobian matrix in (3) can be derived concisely in a straightforward manner as

$$(\dot{\mathbf{g}}_{st}\mathbf{g}_{st}^{-1})^{\vee} = \mathbf{J}_t\,\dot{\boldsymbol{\theta}} \;\Rightarrow\; \mathbf{J}_t = [\boldsymbol{\xi}_1,\, \boldsymbol{\xi}_2,\, \cdots,\, \boldsymbol{\xi}_n] \tag{4}$$

where $(\dot{\mathbf{g}}_{st}\mathbf{g}_{st}^{-1})^{\vee} \in \mathbb{R}^{6\times1}$ represents the generalized velocity of the tip frame. $\boldsymbol{\xi}_i \in \mathbb{R}^{6\times1}$ corresponds to the joint twist in the current configuration, given by

$$\boldsymbol{\xi}_i = \mathrm{Ad}\left(\exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_{i-1}\theta_{i-1})\right)\boldsymbol{\zeta}_i,\;\; i = 1, \cdots, n$$

where $\mathrm{Ad}(\cdot)$ is the adjoint operator of $SE(3)$ (Selig 2005).

Composing the geometric constraint (2) and the equilibrium condition (3), the kinetostatics of the equivalent linkage can be written in a set of nonlinear algebraic equations as

$$\boldsymbol{C}(\boldsymbol{\theta}, \boldsymbol{F}_t) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} \left(\ln(\mathbf{g}_{st}\,\mathbf{g}_t^{-1})\right)^{\vee} \\ \mathbf{K}_{\boldsymbol{\theta}}\,\boldsymbol{\theta} - \mathbf{J}_t^T\,\boldsymbol{F}_t \end{bmatrix} = \boldsymbol{0} \tag{5}$$

where $\boldsymbol{y} = (\ln(\mathbf{g}_{st}\,\mathbf{g}_t^{-1}))^{\vee} \in \mathbb{R}^{6\times1}$ denotes the pose deviation of $\{\mathrm{T}\}$ from the target one. $\boldsymbol{\tau} \in \mathbb{R}^{n\times1}$ relates to the torque deviation of the elastic joints to balance the external load.

The system kinetostatics model (5) corresponds to a set of $n+6$ nonlinear algebraic equations, just in terms of $n+6$ variables. In other words, using the proposed approximation method, the large deformation problem of the constrained elastic beam relates to a deterministic system. Moreover, the system Jacobian matrix of the simultaneous equations can be derived analytically in a straightforward way. Consequently, the Newton-Raphson method can be adopted to accomplish the kinetostatics analysis efficiently.

Using POE formulation, the system Jacobian matrix of the implicit equation (5) can be derived as

$$\boldsymbol{\nabla}_{\boldsymbol{C}} = \begin{bmatrix} \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{y}}{\partial \boldsymbol{F}_t} \\ \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{\theta}} & \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{F}_t} \end{bmatrix} = \begin{bmatrix} \mathbf{J}_t & \mathbf{0} \\ \mathbf{K}_{\boldsymbol{\theta}} - \mathbf{K}_{\mathbf{J}_t} & -\mathbf{J}_t^T \end{bmatrix} \tag{6}$$

where $\mathbf{K}_{\mathbf{J}_t} \in \mathbb{R}^{n\times n}$ relates to the configuration-dependent item of the overall joint stiffness, given by

$$\mathbf{K}_{\mathbf{J}_t} = \frac{\partial(\mathbf{J}_t^T\boldsymbol{F}_t)}{\partial\boldsymbol{\theta}} = \left[\boldsymbol{F}_t^T\frac{\partial\boldsymbol{\xi}_j}{\partial\theta_i}\right]_{j,i} \Rightarrow \frac{\partial\boldsymbol{\xi}_j}{\partial\theta_i} = \begin{cases} \boldsymbol{0}, & i \geq j \\ \mathbf{Z}_i\boldsymbol{\xi}_j, & i < j \end{cases}$$

where $\mathbf{Z}_i = \mathrm{ad}(\hat{\boldsymbol{\xi}}_i)$ is the adjoint representation of $\hat{\boldsymbol{\xi}}_i$.

Taking advantage of the closed-form Jacobian matrix (6), the nonlinear algebraic equation (5) can be iteratively solved by following the update theme

$$\begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{F}_t \end{bmatrix}_{k+1} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{F}_t \end{bmatrix}_{k} - (\boldsymbol{\nabla}_{\boldsymbol{C}})_k^{-1}\,\boldsymbol{C}_k \tag{7}$$

where the subscript 'k' denotes the k^{th} iterative step.

According to (7), the static equilibrium configuration of the equivalent linkage can be determined in an iterative way. Both $\boldsymbol{\theta}$ and $\boldsymbol{F}_t$ converge to stable values quickly, while the function $\boldsymbol{C}$ approaches to zero. The deformed configuration of the constrained beam can then be reconstructed according to the identified joints' displacements. As exhibited in Fig. 2, a numerical example is provided for the studied elastic beam, whose large deformation can be identified efficiently using the Newton-Raphson method.

3 Multi-Point Contact Modeling

Building upon the discretization-based solution for large elastic beam deformations presented in the preceding section, this section extends the analysis by establishing a kinetostatic model in the case of multi-point contact for the deformed elastic beam as a contact sensor. Since large deformations of the studied elastic beam are taken into account, the developed model is not only available for small contact cases, but also works for large-scale interaction.

3.1 Stiffness Modeling of Single-Point Contact

The cases of single-point contact are studied to exhibit the stiffness modeling of the deformed elastic beam touched by an object in intermediate locations. It is worth noting that, in such a case, only small deformations will be generated to the constrained beam due to the contact with objects.

Let $\boldsymbol{F}_c \in \mathbb{R}^{6\times1}$ be an infinitesimal external force exerted at the point C. As shown in Fig. 3, let $\{\mathrm{C}\}$ be the contact frame coincident with C. Due to the existence of reaction force $\boldsymbol{F}_c$, the static equilibrium condition of the equivalent linkage, namely (3), should be updated as

$$\begin{aligned} \boldsymbol{\tau} &= \mathbf{K}_{\boldsymbol{\theta}}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) - \mathbf{J}_t^T(\boldsymbol{F}_t + \delta\boldsymbol{F}_t) - \mathbf{J}_c^T\boldsymbol{F}_c \\ &= \mathbf{K}_{\boldsymbol{\theta}}\delta\boldsymbol{\theta} - \mathbf{J}_t^T\delta\boldsymbol{F}_t - \mathbf{J}_c^T\boldsymbol{F}_c = \boldsymbol{0} \end{aligned} \tag{8}$$

where $\delta\boldsymbol{\theta} = \mathbf{C}_{\boldsymbol{\theta}}(\mathbf{J}_t^T\delta\boldsymbol{F}_t + \mathbf{J}_c^T\boldsymbol{F}_c)$ is the infinitesimal deviation of joint displacements induced by the contact force $\boldsymbol{F}_c$. Here, $\mathbf{C}_{\boldsymbol{\theta}} = \mathbf{K}_{\boldsymbol{\theta}}^{-1}$ represents the joints' compliance matrix. While $\delta\boldsymbol{F}_t$ denotes the extra component of the constraint force. And $\mathbf{J}_c \in \mathbb{R}^{6\times n}$ is the Jacobian matrix related to $\{\mathrm{C}\}$, which can be obtained directly as

$$(\dot{\mathbf{g}}_{sc}\mathbf{g}_{sc}^{-1})^{\vee} = \mathbf{J}_c\,\dot{\boldsymbol{\theta}} \;\Rightarrow\; \mathbf{J}_c = [\boldsymbol{\xi}_1,\, \cdots,\, \boldsymbol{\xi}_{n_c},\, \boldsymbol{0}, \cdots,\, \boldsymbol{0}]$$

where n_c denotes the number of joints in front of $\{\mathrm{C}\}$.

As shown in Fig. 3, in the contact enforced equilibrium configuration, the geometric constraint to the tip frame still holds, such that $\mathbf{J}_t\delta\boldsymbol{\theta} = \boldsymbol{0}$. Then, the extra constraint force can be derived in a linear form of the contact force as

$$\mathbf{C}_t\delta\boldsymbol{F}_t + \mathbf{J}_t\mathbf{C}_{\boldsymbol{\theta}}\mathbf{J}_c^T\boldsymbol{F}_c = \boldsymbol{0} \Rightarrow \delta\boldsymbol{F}_t = -\mathbf{K}_t\,\mathbf{J}_t\,\mathbf{C}_{\boldsymbol{\theta}}\,\mathbf{J}_c^T\,\boldsymbol{F}_c \tag{9}$$

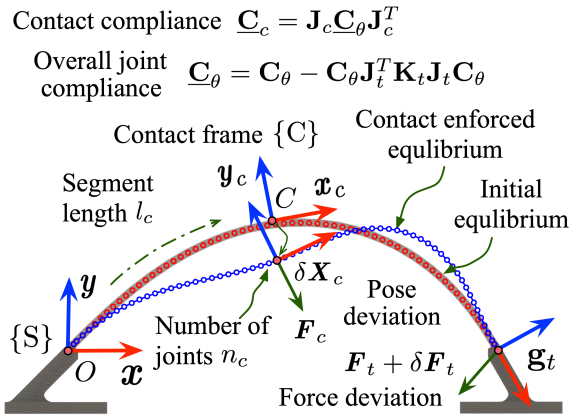

Figure 3. Contact stiffness modeling of the deformed elastic beam.

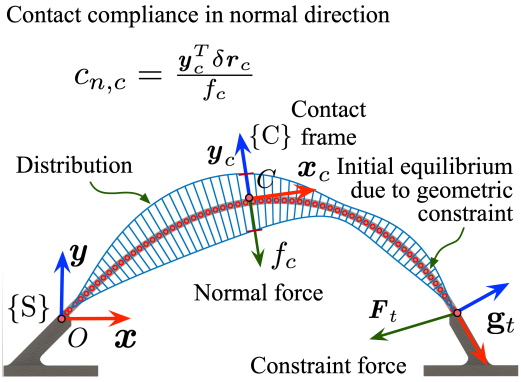

Figure 4. Distribution of normal contact compliance along the deformed beam.

where $\mathbf{K}_t = \mathbf{C}_t^{-1}$ and $\mathbf{C}_t = \mathbf{J}_t\mathbf{C}_\theta\mathbf{J}_t^T \in \mathbb{R}^{6\times6}$ are the stiffness and compliance matrices associated with the tip frame.

Substituting (9) into (8), the deviation of the elastic joints' displacements caused by the additional contact force can then be derived directly as

$$\delta\boldsymbol{\theta} = (\mathbf{C}_\theta - \mathbf{C}_\theta\mathbf{J}_t^T\mathbf{K}_t\mathbf{J}_t\mathbf{C}_\theta)\mathbf{J}_c^T\boldsymbol{F}_c = \underline{\mathbf{C}}_\theta\mathbf{J}_c^T\boldsymbol{F}_c \tag{10}$$

which $\underline{\mathbf{C}}_\theta = \mathbf{C}_\theta - \mathbf{C}_\theta\mathbf{J}_t^T\mathbf{K}_t\mathbf{J}_t\mathbf{C}_\theta \in \mathbb{R}^{n\times n}$ denotes the overall compliance matrix of the linkage's elastic joints, including the configuration-dependent item $-\mathbf{C}_\theta\mathbf{J}_t^T\mathbf{K}_t\mathbf{J}_t\mathbf{C}_\theta$.

Eventually, the force-deflection behavior of the deformed beam at the contact frame {C} can be obtained as

$$\delta\boldsymbol{X}_c = \mathbf{J}_c\delta\boldsymbol{\theta} = \mathbf{J}_c\underline{\mathbf{C}}_\theta\mathbf{J}_c^T\boldsymbol{F}_c = \underline{\mathbf{C}}_c\boldsymbol{F}_c \;\Rightarrow\; \boldsymbol{F}_c = \underline{\mathbf{K}}_c\delta\boldsymbol{X}_c \tag{11}$$

where $\underline{\mathbf{C}}_c = \mathbf{J}_c\underline{\mathbf{C}}_\theta\mathbf{J}_c^T$ and $\underline{\mathbf{K}}_c = \underline{\mathbf{C}}_c^{-1}$ are the corresponding contact compliance and stiffness matrices relating the force $\boldsymbol{F}_c$ and the deformation $\delta\boldsymbol{X}_c$ directly.

From (10), it is known that the overall compliance matrix $\underline{\mathbf{C}}_\theta$ only depends on the Jacobian $\mathbf{J}_t$ associated with the tip frame. Here, the compliance matrix $\mathbf{C}_\theta$ is constant, once the approximation to the elastic beam is achieved. Consequently, $\underline{\mathbf{C}}_\theta$ can be completely determined according to the geometric constraint to the elastic beam. Thus, the contact compliance (stiffness) about the location {C} on the constrained elastic beam can be determined uniquely through the congruence mapping (11) via the corresponding Jacobian $\mathbf{J}_c$.

Moreover, the local Jacobian $\mathbf{J}_c$ simply corresponds to the first n_c columns of $\mathbf{J}_t$. Therefore, the contact compliance (stiffness) matrix for a given position can be represented in closed-form, once the constrained deformation of the slender elastic beam is specified using the proposed method.

With the explicit expression of the contact compliance, the enforced deformation can be calculated directly according to the linearized force-deflection relationship (11), with the knowledge of contact force. Reversely, the corresponding interaction force could also be determined as long as the deformation of the contact frame can be specified in a certain way. It is worth noting that, the established stiffness model is only valid for the scenario of small contact forces, as well as the corresponding induced deformations. Therefore, in this paper, it is only used to pinpoint the initial contact condition for the proposed contact sensing methodology.

In the studied case, the contact wrench $\boldsymbol{F}_c \in \mathbb{R}^{6\times1}$ will degenerate to a three-dimensional planar wrench, so does the deformation twist $\delta\boldsymbol{X}_c$. Meanwhile, the contact friction generated between the object and the beam is neglected in our model. In other words, only the normal stiffness affects the force-deflection behavior of the constrained elastic beam. The unit-force induced deformations are illustrated in Fig. 4, to exhibit the distribution of the contact compliance normal to the deformed beam. From the result, it can be seen that the contact stiffness/compliance varies at different locations of the geometrically-constrained elastic beam.

3.2 *Kinetostatic Modeling of Multi-Point Constraints*

As addressed in Sec. 3.1, the contact stiffness model (11) is only valid for small deformations. This section presents the kinetostatics modeling and analysis of the deformed elastic beam undergoing large-scale contact at multiple locations. To simplify the discussion, but without loss of generality, the two-point contact scenario is provided to elaborate the modeling for the large-deformation problems.

As shown in Fig. 5, two circular objects are prescribed for the deformed elastic beam to contact. Let $\{C_1\}$ and $\{C_2\}$ be two floating frames along the beam. Their poses can then be specified in terms of l_1 and l_2, respectively, which correspond to the curved lengths from the proximal end to the origins. In this case, except for (2), additional geometric constraints need to be imposed at the contact frames as

$$\begin{cases} \mathbf{g}_{s1}(\boldsymbol{\theta}) = \exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_{n_1}\theta_{n_1})\,\mathbf{g}_{s1,0} = \mathbf{g}_1 \\ \mathbf{g}_{s2}(\boldsymbol{\theta}) = \exp(\hat{\boldsymbol{\zeta}}_1\theta_1)\cdots\exp(\hat{\boldsymbol{\zeta}}_{n_2}\theta_{n_2})\,\mathbf{g}_{s2,0} = \mathbf{g}_2 \end{cases} \tag{12}$$

where $\mathbf{g}_{s1}$ and $\mathbf{g}_{s2}$ denote the relative poses of $\{C_1\}$ and $\{C_2\}$, respectively, with respect to {S}. Similarly, $\mathbf{g}_{s1,0}$ and $\mathbf{g}_{s2,0}$ are the initial poses at $\boldsymbol{\theta} = \boldsymbol{0}$, which can be specified in terms of l_1 and l_2, as indicated in Fig. 5. Here, n_1 and n_2 represent the numbers of joints in front of $\{C_1\}$ and $\{C_2\}$, respectively. While $\mathbf{g}_1 \in SE(3)$ and $\mathbf{g}_2 \in SE(3)$ are the target poses for the contact frames floating along the beam.

As well, the equilibrium condition in the linkage's joint space should be updated by taking the contact forces into account. Then, the static equilibrium condition of the elastic joints, namely (3), can be rewritten as

$$\boldsymbol{\tau} = \mathbf{K}_\theta\boldsymbol{\theta} - \mathbf{J}_t^T\boldsymbol{F}_t - \mathbf{J}_1^T\boldsymbol{F}_1 - \mathbf{J}_2^T\boldsymbol{F}_2 = \boldsymbol{0} \tag{13}$$

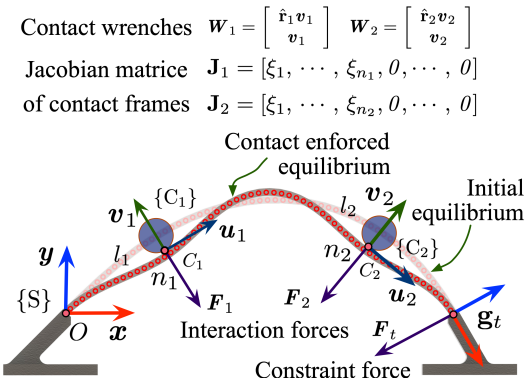

Figure 5. Contact model of two-point constraints in intermediate location.

where $\mathbf{J}_1 = [\boldsymbol{\xi}_1, \cdots, \boldsymbol{\xi}_{n_1}, \boldsymbol{0}, \cdots, \boldsymbol{0}] \in \mathbb{R}^{6\times n}$ and $\mathbf{J}_2 = [\boldsymbol{\xi}_1, \cdots, \boldsymbol{\xi}_{n_2}, \boldsymbol{0}, \cdots, \boldsymbol{0}] \in \mathbb{R}^{6\times n}$ are associated with $\{\mathrm{C}_1\}$ and $\{\mathrm{C}_2\}$, respectively.

As indicated in Sec. 3.1, frictionless contact is considered in the proposed kinetostatics model. Thus, the contact force $\boldsymbol{F}_i \in \mathbb{R}^{6\times 1} (i = 1,2)$ in (13) is assumed to exert at the origin of $\{\mathrm{C}_i\}$ and along the normal direction, namely C_i and $\boldsymbol{v}_i$-axis, respectively. Then, the interaction force can be represented in the general form of 6×1 wrenches, as

$$\boldsymbol{F}_i = f_i \begin{bmatrix} \hat{\mathbf{r}}_i \boldsymbol{v}_i \\ \boldsymbol{v}_i \end{bmatrix} = f_i \boldsymbol{W}_i, \quad i = 1,2 \tag{14}$$

where $f_i, i = 1,2$ are the magnitudes of the contact forces $\boldsymbol{F}_i$. $\boldsymbol{W}_i \in \mathbb{R}^{6\times 1}$ corresponds to the unit wrench. Here, $\boldsymbol{r}_i \in \mathbb{R}^{3\times 1}$ is the contact position, namely the origin C_i, as drawn in Fig. 5. $\boldsymbol{v}_i \in \mathbb{R}^{3\times 1}$ denotes the unit direction vector of $\boldsymbol{F}_i$, which is just normal to the deformed elastic beam at C_i. Hence, the unit wrench $\boldsymbol{W}_i$ can be specified uniquely according to the pose of the contact frame $\{\mathrm{C}_i\}$.

Combining the geometric constraints (5) and (12), and the equilibrium condition (13) together, the kinetostatics model of the elastic beam in multiple contact can be represented in the general form of nonlinear algebraic equations as

$$\boldsymbol{C}(\boldsymbol{\theta}, \boldsymbol{F}_t, \boldsymbol{l}, \boldsymbol{f}) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \\ \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \end{bmatrix} = \begin{bmatrix} (\ln(\mathbf{g}_{st}\, \mathbf{g}_t^{-1}))^\vee \\ \mathbf{K}_{\boldsymbol{\theta}}\, \boldsymbol{\theta} - \mathbf{J}_t^T \boldsymbol{F}_t - \sum_{i=1}^{2} \mathbf{J}_i^T \boldsymbol{F}_i \\ (\ln(\mathbf{g}_{s1}\, \mathbf{g}_1^{-1}))^\vee \\ (\ln(\mathbf{g}_{s2}\, \mathbf{g}_2^{-1}))^\vee \end{bmatrix} \tag{15}$$

where $\boldsymbol{l} = [l_1, l_2]^T$ and $\boldsymbol{f} = [f_1, f_2]^T$ are the additional variables required to be identified in the contact problems with multiple intermediate constraints.

Comparing with (5), the kinetostatic model of multi-point contact (15) has four extra variables to be identified, namely $\boldsymbol{l}$ and $\boldsymbol{f}$. On the other hand, the intermediate constraints $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ introduce four extra independent equations. Consequently, the equilibrium configuration of the deformed elastic beam in contact with objects at multiple locations can be determined by solving these nonlinear algebraic equations. It should be noted that, the deformed elastic beam is in contact with the objects at $\{\mathrm{C}_1\}$ and $\{\mathrm{C}_2\}$. More specifically, the two frames should be locally coincident with the normal directions of the objects at the contact locations. In the studied case, target objects in circular shapes are used, such that the constraints $\boldsymbol{y}_i$ could be instantiated as two-dimensional ones. Therefore, the extra geometric constraints to the deformed beam can be represented exactly as $\|\boldsymbol{r}_i - \boldsymbol{r}_{O_i}\| = r_0$ and $(\boldsymbol{r}_i - \boldsymbol{r}_{O_i})^T \boldsymbol{u}_i = 0$ $(i = 1,2)$. Here, $\boldsymbol{r}_{O_i}$ and r_0 correspond to the positions and radius of the circular objects.

Similar to (6), the system Jacobian matrix of the nonlinear algebraic equations (15) with respect to all those unknown variables can be derived in a concise form as

$$\boldsymbol{\nabla}_{\boldsymbol{C}} = \begin{bmatrix} \mathbf{J}_t & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{K}_{\boldsymbol{\theta}} - \mathbf{K}_{\mathbf{J}_t} - \sum_{i=1}^{2} \mathbf{K}_{\mathbf{J}_i} & -\mathbf{J}_t^T & \mathbf{J}_{\boldsymbol{\tau},l} & \mathbf{J}_{\boldsymbol{\tau},f} \\ \mathbf{J}_1 & \mathbf{0} & \mathbf{J}_{1,l} & \mathbf{0} \\ \mathbf{J}_2 & \mathbf{0} & \mathbf{J}_{2,l} & \mathbf{0} \end{bmatrix} \tag{16}$$

where $\boldsymbol{y}$ is independent of $\boldsymbol{l}$ and $\boldsymbol{f}$. The initial poses $\mathbf{g}_{s1,0}$ and $\mathbf{g}_{s2,0}$ can be uniquely determined by $\boldsymbol{l}$, which also influence the contact frames and the interaction forces. Likewise, the block elements of $\boldsymbol{\nabla}_{\boldsymbol{C}}$ can be obtained in a straightforward manner. Moreover, all those in (6), including the item $\mathbf{J}_t$, $\mathbf{K}_{\boldsymbol{\theta}}$ and $\mathbf{K}_{\mathbf{J}_t}$, can be reused in this contact kinetostatics model with multiple constraints. The details of derivation of the rest block entries are given in Appendix.

Owing to the analytical system Jacobian matrix (16), the kinetostatic model of the pre-deformed elastic beam, that is geometrically constrained by objects at multiple points, can also be solved efficiently using the Newton-Raphson method. Likewise, the update theme of the iterative algorithm can be implemented in the same way as (7) does. The terminate condition includes not only the geometric constraint to the tip frame, viz. $\boldsymbol{y}$, but also those induced by the contact objects, namely $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$. Meanwhile, the static balance of the elastic joints, namely $\boldsymbol{\tau}$, should also be satisfied to fulfill both the geometric constraints and the static balance.

As a result, the contact-related equilibrium configuration of the deformed elastic beam can be constructed according to the obtained joint displacement $\boldsymbol{\theta}$. As demonstrated in Fig. 5, an instance is provided to manifest the enforced deformation of the elastic beam in multi-point contact. Meanwhile, the exact contact locations can be obtained from the equilibrium poses of $\{\mathrm{C}_i\}$ through $\boldsymbol{l}$, so are the unit wrench $\boldsymbol{W}_i$. Together with $\boldsymbol{f}$, the interaction forces generated between the elastic beam and the objects in contact can be determined. Hence, the problem of multi-point contact of the studied elastic beam can be fully solved using the proposed method. It is worth noting that, the kinetostatic model (15) is not only available for small deformation contact, but also works for large-scale interaction with objects. And it will serve as the mechanics fundamental of the contact sensing methodology proposed in this paper.

4 Modeling of Contact Sensing Methodology

Based on the kinetostatic model that established in Sec. 3, the present section proceeds to formulate a methodology for multi-point contact sensing that exploits the constrained elastic beam as an intermediary between the robot and its environment. Two distinct models of contact sensing are particularly developed for both small-deformation contact

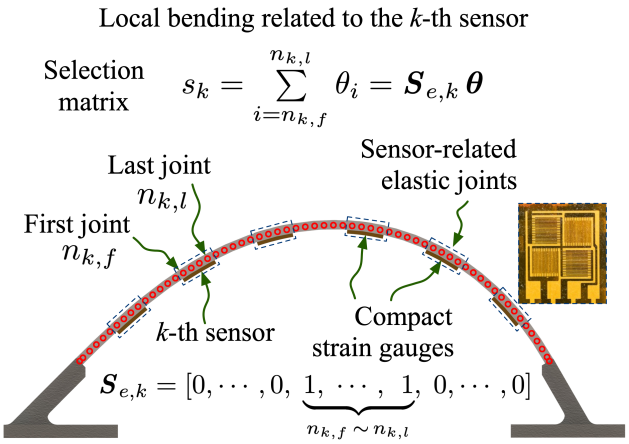

Figure 6. Deployment of the strain gauges for local bending measuring.

and large-scale interaction, respectively. In the proposed sensing method, the compact strain gauges (integrated on FPCB) introduced in Sec. 2.1, are employed to measure the bending information of the deformed beam in the locations where the deflection sensors are deployed.

4.1 Sensing Model for Small-Deformation Contact

In the scenario of small contact, the enforced deformation of the constrained elastic beam could be linearly related to the corresponding reaction force via the contact compliance (11). As addressed in Sec. 2, the proposed contact sensing method relies on the knowledge of the beam's local bending measured through the developed strain gauges, rather than the direct deformation that generated at the contact positions. Hence, the overall compliance of the elastic joints, namely $\underline{\mathbf{C}}_{\boldsymbol{\theta}}$ in (10), is used to construct the linear mapping between the deviation of joint displacements and the small contact force. As a result, the location of contact and the magnitude of interaction force can be predicted in terms of the bending information gathered by the locally-deployed gauges.

In the proposed model, m strain gauges are deployed to the studied elastic beam to collect the information of local bending enforced by external contact, as illustrated in Fig. 6. According to the linearity of the strain gauge's bending and resistance, extra equations can then be imposed by virtue of the strain gauges' readings, as

$$\mathbf{S}_e\delta\boldsymbol{\theta} - \delta\boldsymbol{s} = \boldsymbol{0} \tag{17}$$

where $\delta\boldsymbol{\theta} = \boldsymbol{\theta} - \boldsymbol{\theta}_0$ represents the small deviation of the joint displacements, with respect to the pre-deformed equilibrium configuration $\boldsymbol{\theta}_0$, caused by the contact force. $\delta\boldsymbol{s} = \boldsymbol{s} - \boldsymbol{s}_0$ is the corresponding deviation of the bending angles relating to the deployed strain gauges. Here, $\boldsymbol{s} \in \mathbb{R}^{m\times 1}$ represents the vector of the gauges' readings and $\boldsymbol{s}_0$ is the one associated with $\boldsymbol{\theta}_0$. While $\mathbf{S}_e \in \mathbb{R}^{m\times n}$ represents the selection matrix relating the joint variables to the signal readings of the used strain gauges. Once the deployment of the strain gauges is specified, the matrix $\mathbf{S}_e$ can be obtained directly as

$$\mathbf{S}_e = \begin{bmatrix} \boldsymbol{S}_{e,1} \\ \vdots \\ \boldsymbol{S}_{e,k} \\ \vdots \\ \boldsymbol{S}_{e,m} \end{bmatrix} \Rightarrow \boldsymbol{S}_{e,k} = [0,\cdots,0,\underbrace{1,\cdots,1}_{n_{k,f}\sim n_{k,l}},0,\cdots,0] \in \mathbb{R}^{1\times n}$$

where $n_{k,f}$ and $n_{k,l}$ $(k = 1,\cdots,m)$ are the first and last joints the k^{th} strain gauge is related, as shown in Fig. 6.

The compliance model of single-point contact (10) can be naturally extended to multi-point situations. As a result, the contact-induced deviation of the joint variable due to the two interaction forces can be represented as

$$\delta\boldsymbol{\theta} = \underline{\mathbf{C}}_{\boldsymbol{\theta}}\left(\mathbf{J}_1^T\boldsymbol{F}_1 + \mathbf{J}_2^T\boldsymbol{F}_2\right) = \underline{\mathbf{C}}_{\boldsymbol{\theta}}\sum_{i=1}^{2}\mathbf{J}_i^T\boldsymbol{W}_i f_i = \mathbf{T}\boldsymbol{f} \tag{18}$$

which $\boldsymbol{F}_i$, $i = 1,2$ and $\mathbf{J}_i$, $i = 1,2$ denote the contact force and the corresponding Jacobian defined in (13). $\boldsymbol{W}_i$ and f_i denote the unit wrench and the magnitude of $\boldsymbol{F}_i$, respectively. While $\mathbf{T} = [\boldsymbol{T}_1, \boldsymbol{T}_2] \in \mathbb{R}^{n\times 2}$ with the columns $\boldsymbol{T}_i = \underline{\mathbf{C}}_{\boldsymbol{\theta}}\mathbf{J}_i^T\boldsymbol{W}_i$.

Substituting (18) into (17), the sensing model for two-point contact with small deformation can be rewritten as

$$\mathbf{S}_e\delta\boldsymbol{\theta} = \mathbf{S}_e\mathbf{T}\boldsymbol{f} = \delta\boldsymbol{s} \tag{19}$$

In model (19), the vector $\delta\boldsymbol{s}$ can be obtained via the signal readings of the strain gauges. Meanwhile, the matrix $\mathbf{T}$ can be uniquely determined according to $\boldsymbol{l}$, which relates to the contact locations. Together with $\boldsymbol{f}$, there exist totally four independent variables need to be identified in this sensing model. Therefore, at least four strain gauges are required to estimate the locations and forces in this two-point contact scenario. In this paper, eight strain gauges, viz. $m = 8$, are deployed onto the elastic beam in a set of discrete locations. Thus, enough bending information can be collected to fully identify both the location and the magnitude of the contact forces. In the proposed contact sensing method, a two-step algorithm is developed to solve the non-linear equations (18) for $\boldsymbol{l}$ and $\boldsymbol{f}$, consecutively.

First, the well-known unit-load method (Udoeyo 2019) is employed to identify the locations of contact. In this step, the contact forces are assumed to be exerted in the middle of the discretized small segments. For each pair of $\{i, j\}$-joint, the unit wrenches and the corresponding Jacobian matrices can be obtained directly as

$$\begin{cases} \boldsymbol{W}_{m_i} = \begin{bmatrix} \hat{\mathbf{r}}_{m_i}\boldsymbol{v}_{m_i} \\ \boldsymbol{v}_{m_i} \end{bmatrix}, \ \mathbf{J}_{m_i} = [\boldsymbol{\xi}_1,\cdots,\boldsymbol{\xi}_i,\boldsymbol{0},\cdots,\boldsymbol{0}] \\ \boldsymbol{W}_{m_j} = \begin{bmatrix} \hat{\mathbf{r}}_{m_j}\boldsymbol{v}_{m_j} \\ \boldsymbol{v}_{m_j} \end{bmatrix}, \ \mathbf{J}_{m_j} = [\boldsymbol{\xi}_1,\cdots,\boldsymbol{\xi}_j,\boldsymbol{0},\cdots,\boldsymbol{0}] \end{cases} \tag{20}$$

where $\boldsymbol{r}_{m_i}$ and $\boldsymbol{r}_{m_j}$ are the position vectors of middle point M_i and M_j, as drawn in Fig. 7. $\boldsymbol{v}_{m_i}$ and $\boldsymbol{v}_{m_j}$ represent their normal directions. It is worth noting that, all those variables are related to the deformed configuration of the elastic beam constrained at tip frame. Therefore, they can be determined uniquely by the pair $\{i, j\}$ for $i, j = 1,\cdots,n$.

According to (20), the two-point contact sensing model (19) can be rewritten in a concise form as

$$\boldsymbol{C}_s(i,j) = \boldsymbol{\Gamma}_i f_i + \boldsymbol{\Gamma}_j f_j + \boldsymbol{\Gamma}_0 = \boldsymbol{0} \tag{21}$$

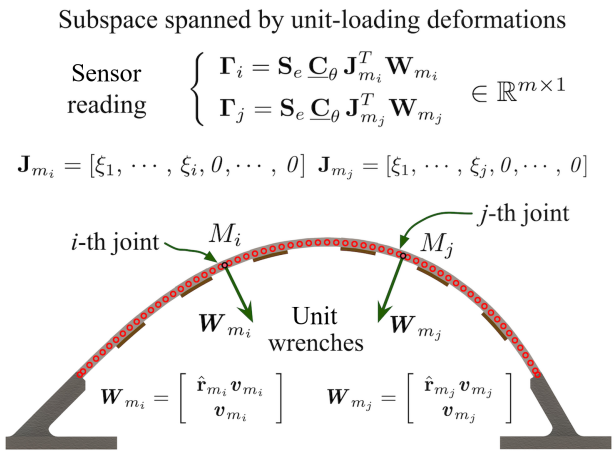

Figure 7. Unit-load method of deflection for the deformed elastic beam.

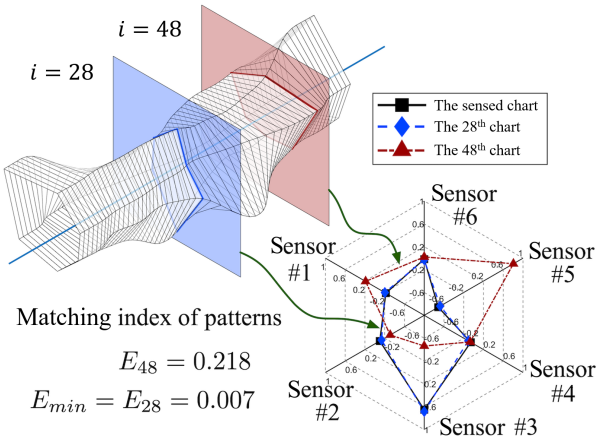

Figure 8. Contact identification using Gram determinate.

where $\Gamma_i = \mathbf{S}_e \underline{\mathbf{C}}_{\boldsymbol{\theta}} \mathbf{J}_{m_i}^T \boldsymbol{W}_{m_i}$ and $\Gamma_j = \mathbf{S}_e \underline{\mathbf{C}}_{\boldsymbol{\theta}} \mathbf{J}_{m_j}^T \boldsymbol{W}_{m_j}$ denote the $m \times 1$ vectors that relate the unit loads at $\{i, j\}$ to the bending of the strain-gauge sensors. And $\Gamma_0 = -\boldsymbol{\delta s}$ for simplification in representation.

From (21), it is known that the enforced deformation of the elastic beam can be realized by two forces exerted at the i^{th} and j^{th} joints, if the gauges' reading Γ_0 simply corresponds to a linear combination of Γ_i and Γ_j. As a consequence, the model for contact sensing can be reformulated to find an appropriate pair of $\{i, j\}$, such that the three vectors Γ_0, Γ_i and Γ_j are just linearly dependent. It should be noted that, in this model, the locations of contact are assumed in the middle of the segments, rather than identified precisely. In other words, exact solutions to the sensing model (21) can not be guaranteed all the time.

For the above problem, the concept of Gram determinate (Horn and Johnson 2012) is adopted to evaluate the linear dependency of Γ_0 relative to Γ_i and Γ_j, given by

$$E_{i,j} = G(\bar{\Gamma}_0, \bar{\Gamma}_i, \bar{\Gamma}_j) = \begin{vmatrix} \bar{\Gamma}_0^T \bar{\Gamma}_0 & \bar{\Gamma}_0^T \bar{\Gamma}_i & \bar{\Gamma}_0^T \bar{\Gamma}_j \\ \bar{\Gamma}_i^T \bar{\Gamma}_0 & \bar{\Gamma}_i^T \bar{\Gamma}_i & \bar{\Gamma}_i^T \bar{\Gamma}_j \\ \bar{\Gamma}_j^T \bar{\Gamma}_0 & \bar{\Gamma}_j^T \bar{\Gamma}_i & \bar{\Gamma}_j^T \bar{\Gamma}_j \end{vmatrix} \tag{22}$$

where $\bar{\Gamma}_0$, $\bar{\Gamma}_i$ and $\bar{\Gamma}_i$ are the normalized vectors of Γ_0, Γ_i and Γ_j, respectively.

From a geometric point of view, the Gram determinate (22) can be considered as an assessment of closeness from the vector $\bar{\Gamma}_0$ to the two-dimensional subspace spanned by $\{\bar{\Gamma}_i, \bar{\Gamma}_j\}$. Hence, there are totally $\frac{1}{2}n^2 - n$ subspace can be constructed for all $\{i, j\}$ pairs in advance (Here, $i \neq j$). For each set of strain gauge readings $\bar{\Gamma}_0$, its closeness to all those subspaces can be calculated directly according to (22). Then, the one with the minimum determinate, denoted $\{i^*, j^*\}$, is selected as a best estimation to the contact sensing model. Accordingly, the points M_{i^*} and M_{j^*} just correspond to the locations of contact, from which an approximation to the unknown variable $\boldsymbol{l}_*$ can be determined directly.

Once the locations of contact are specified, the magnitudes of the interaction forces can be obtained by means of solving the linear equation (21) for the unknown $\boldsymbol{f}$ as

$$\boldsymbol{f}_* = [\Gamma_{i^*}, \Gamma_{j^*}]^+ \boldsymbol{\delta s} \tag{23}$$

where Γ_{i^*} and Γ_{j^*} are related to M_{i^*} and M_{j^*}, respectively. And $(\cdot)^+$ represents the pseudo-inverse.

Using the proposed contact sensing method, the locations and magnitudes of the contact forces can be simultaneously identified in a straightforward way. It should be noted that, the solutions obtained in this step are just approximate ones, since the contact is supposed to occur at the middle points of the discretized segments. Specifically, all locations within the same segment are uniformly centralized to the identical point. Fortunately, the length of the discretized segments is relatively small. In the proposed sensing method, a $300mm$ elastic beam is discretized into 60 segments ($5mm$ for each). For further improvement, this approximate estimation can act as an initial guess for the exact sensing model which can be solved using the iterative algorithms. This will be intensively discussed in the following section.

For ease of understanding, an example of single-point case is provided to elaborate the determination of contact location. In such a scenario, the pattern of local bending corresponding to the unit load exerted at the i^{th} segment can be represented in an m-dimensional radar chart, as drawn in Fig. 8. Thus, the linear dependency between the readings of the strain gauges and those associated with the i^{th} unit load can be evaluated by the similarity of the corresponding two radar charts. In a sense, it simply relates to the reduced Gram determinate $E_i = G(\bar{\Gamma}_0, \bar{\Gamma}_i)$. Consequently, the location of contact can be identified by comparing the gauges' readings to all potential patterns of local bending and specifying the most similar one. An instance is exhibited in Fig. 8 to demonstrate the result in an intuitive way. From the figure, it is known that the exact contact location can be identified efficiently.

Besides, the linearized sensing model developed in this section is only available for contact of small deformations. Otherwise, the nonlinear effect of motion/force transmission would influence the accuracy of prediction. This problem will be conquered in the following section, based on the kinetostatic model of large-deformation contact. It can also be adopted to improve the prediction results obtained from the method relying on contact compliance.

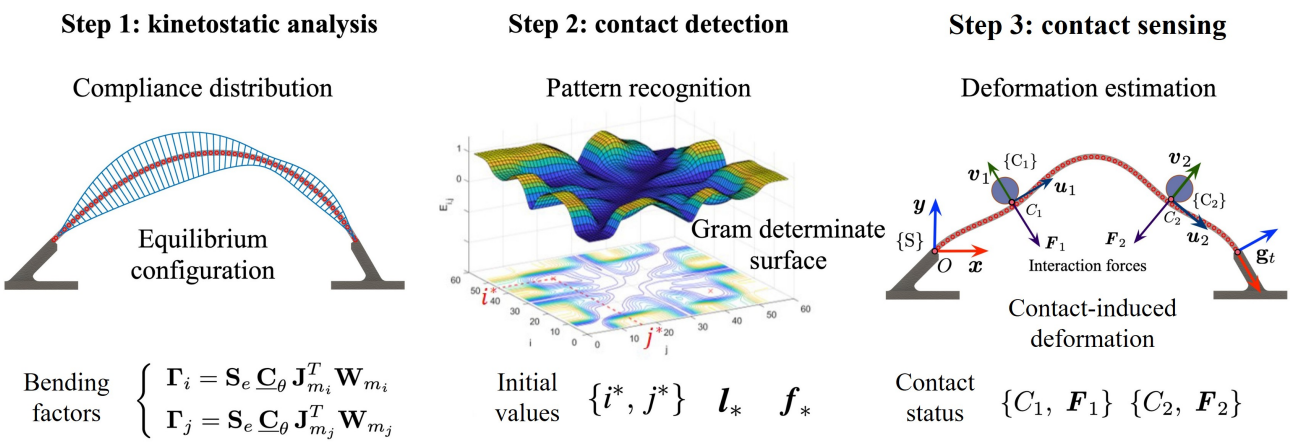

Figure 9. Flow chart of the proposed contact sensing methodology.

4.2 Sensing Model for Large-Scale Interactions

As addressed in Sec. 3.1, the model of contact compliance relies on the assumption of small deformation. In this section, the kinetostatic model of the constrained elastic beam (15) is modified particularly for sensing the large-scale interaction. Likewise, local bending that gathered by the strain gauges is imposed to release the geometric constraints caused by the objects. An example of two-point interaction is exemplified to illustrate the contact sensing methodology.

Different from the kinetostatic model (15), the geometric constraints due to the contact objects are not given in advance in the sensing model. Instead, the joint displacements related to the strain gauges can be acquired as extra constraints to the sensing model. Therefore, the contact sensing model of the constrained elastic beam for large-scale interactions can be represented as

$$\boldsymbol{S}(\boldsymbol{\theta},\boldsymbol{F}_t,\boldsymbol{l},\boldsymbol{f}) = \begin{bmatrix} \boldsymbol{y} \\ \boldsymbol{\tau} \\ \boldsymbol{b} \end{bmatrix} = \begin{bmatrix} (\ln(\mathbf{g}_{st}\mathbf{g}_t^{-1}))^{\vee} \\ \mathbf{K}_{\boldsymbol{\theta}}\boldsymbol{\theta} - \mathbf{J}_t^T\boldsymbol{F}_t - \sum_{i=1}^{2}\mathbf{J}_i^T\boldsymbol{F}_i \\ \mathbf{S}_e\boldsymbol{\theta} - \boldsymbol{s} \end{bmatrix} \tag{24}$$

where $\boldsymbol{b} = \mathbf{S}_e\boldsymbol{\theta} - \boldsymbol{s} \in \mathbb{R}^{m\times 1}$ represents the extra constraints to the joint displacements at the locations where the strain gauges are deployed.

In the contact sensing model (24), there are totally $n+m+6$ equations in terms of $n+10$ unknown variables. Moreover, it inherits all variables and most of equations from the kinetostatic one (15). And the extra sensor-induced constraint $\boldsymbol{b}$ is linearly related to the joint displacements. Hence, the corresponding system Jacobian matrix can be derived readily as

$$\boldsymbol{\nabla}_S = \begin{bmatrix} \mathbf{J}_t & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{K}_{\boldsymbol{\theta}} - \mathbf{K}_{\mathbf{J}_t} - \sum_{i=1}^{2}\mathbf{K}_{\mathbf{J}_i} & -\mathbf{J}_t^T & \mathbf{J}_{\boldsymbol{\tau},l} & \mathbf{J}_{\boldsymbol{\tau},f} \\ \mathbf{S}_e & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{25}$$

where all the block elements can be reused from (16).

As mentioned in the above, eight strain-gauge sensors are used in the presented model, namely $m = 8$. Therefore, the sensing model (24) corresponds to a set of over-constrained equations, so that the least square solution can be acquired as the best estimation for the enforced large-scale deformations, as well as the locations and magnitudes of the contact points. Owing to the closed-form system Jacobian (25), the contact sensing model can be solved efficiently in an iterative way using the Newton-Raphson method.

In the developed model, the equilibrium configuration of the constrained elastic beam can be determined once and for all. The identified variable $\boldsymbol{\theta}$ and $\boldsymbol{F}_t$ can serve as an initial estimation for the exact contact sensing model to improve the convergence of the iterative algorithm. Further, the strain gauges are measuring the elastic beam's structural deflection at a high frequency and hence the solving algorithm is able to run at a high rate. As a result, during the sensing process, the deviation of contact-induced deformation between two adjacent steps will be relatively small. Owing to the closed-form Jacobian, the computational efficiency of the proposed sensing method can then be guaranteed to realize the contact perception in real-time, which is crucial for the force-based interactive manipulation.

In the proposed contact sensing method, both the small-deformation model and the large-scale interaction one can be employed to perceive the contact with environments. First, the initial equilibrium state of the constrained elastic beam that is determined in Sec. 2.2, is adopted as a reference to detect the existence of contact. By recalling the small-contact sensing model in Sec. 4.1, the locations and magnitudes of contact can then be identified straightforwardly at the very beginning of contact. And the results obtained in this step directly correspond to the estimated state of contact, if the contact-induced deformation is small. Otherwise, it could be serving as the initial guess of the locations of contact, and as well, the magnitudes of force, for the large-scale interaction model, as presented in Sec. 4.2. Since the initial guesses for the variables in (24) can be specified in a deterministic way, rather than just randomly generated, the convergence of the iterative algorithm can be improved.

To illustrate the developed contact sensing method in an intuitive way, a graphic flow-chart is provided in Fig. 9, to demonstrate the process of identifying the status of contact via the medium of the deformed elastic beam. As indicated in the above, all these procedures can be accomplished readily using the kinetostatics analysis method that proposed in this paper. This ensures the effectiveness and efficiency

of the proposed contact sensing methodology, even for the scenarios of multi-point contact with large deformation. It is worth noting that, the contact sensing algorithm is running continuously, such that only those incremental deformations between the consecutive steps need to be taken into account. Consequently, a real-time estimation can be realized for the proposed contact sensing methodology.

5 Validation Experiments

The preceding derivations establish the theoretical feasibility of the proposed approach. In this section, an experimental apparatus is developed for the validation of the proposed contact sensing methodology. As shown in Fig. 10, a slender carbon-fiber sheet is used as the elastic beam to construct the contact sensor. Meanwhile, a robotic arm with an adjustable end-effector is employed to make contact with the deformed beam in different ways. A commercial force sensor and laser tracker are adopted to measure the contact force and position. Various experiments are carried out to validate the effectiveness of the developed contact sensor for the perception of multi-point contact.

5.1 Experimental Setup

As demonstrated in Fig. 10, a comprehensive experimental apparatus is built for the validation of the proposed contact sensing methodology. A $300\,mm$-long carbon-fiber thin sheet with a size of $14\,mm \times 0.5\,mm$ cross-section, is introduced as the slender elastic beam for contact sensing. The carbon-fiber sheet is stationarily fixed to an optical table at both the proximal and the distal ends via aluminum mounting bases. In the developed prototype, eight strain gauges are deployed on the backside of the beam to measure the local bending at a series of discrete positions. A robotic arm is used to perform interaction with the deformed beam via an adjustable two-pulley end-effector. Hence, the positions of the two pulleys can be controlled separately to interact with the elastic beam in a flexible way. A laser tracker is used to capture the exact positions of the interaction pulleys precisely. In addition, a force/torque sensor is employed to measure the contact force reacted between the elastic beam and the frictionless pulleys. Besides, the information of the beam's local bending can be collected by the strain gauges via the particularly designed sampling circuit, as introduced in Sec. 2.1. Details on the characterizations of the particularly fabricated strain gauges and the corresponding sampling system are provided in the supplementary material.

To make the developed contact sensor be able to run in real-time, an efficient communication framework is built up to connect the hardwares, softwares, and algorithms together, where the SoftBUS technique is used to integrate all modules, including the cooperative robot, the force/torque sensor, the position measuring system, the strain gauges, and the beam's kinetostatics. Here, the algorithm for kinetostatics analysis is implemented in C++ environment, so are the APIs for the robot controller, the interface of F/T sensor, and the position measuring system, as well as the sampling circuit of the strain gauges. Hence, the developed sensing system can run efficiently at the rate of about $30\,Hz$ for contact sensing of multi-point interaction. Further, the validation experiments can be carried out in an autonomous way, for interaction

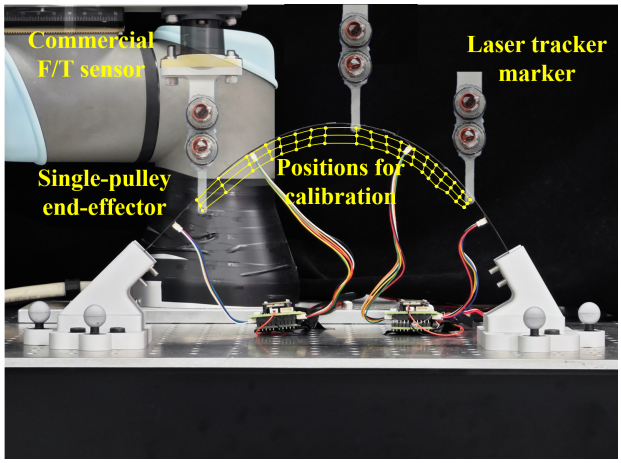

Figure 10. Kinetostatic calibration of the contact sensor.

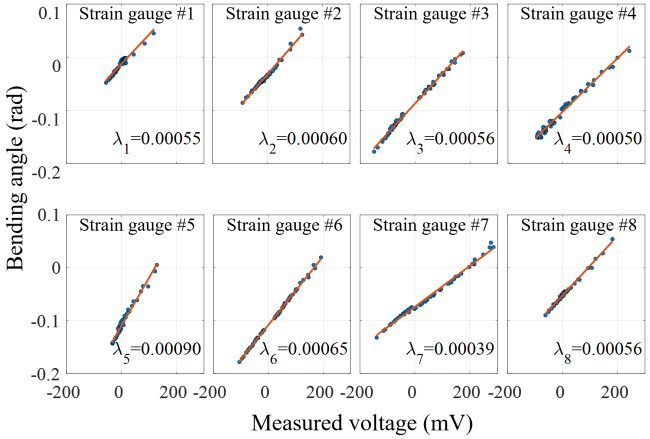

Figure 11. Identification of the strain gauges' properties.

applying, position capturing, force measuring, and contact estimation. This also ensures the real-time feedback control of the robotic systems that integrate the developed contact sensors, for interactive manipulation.

5.2 Kinetostatic Calibration of the Developed Contact Sensor

For the developed prototype, the accuracy of contact sensing is affected by not only the kinetostatic model and mechanical property of the elastic beam, but also the sensitivity ratios of the strain gauges and the deploying position errors. Thus, kinetostatic calibration is necessary for the sensor-embedded elastic beam to identify the interaction-deformation-sensing relation exactly. In this section, experiments of single-point contact are carried out to acquire sufficient information for kinetostatic calibration.

As shown in Fig. 10, a single-pulley end-effector is used to apply contact with the deformed elastic beam at a series of prescribed locations actively. In the process, a laser tracker and a commercial F/T sensor are used to measure the exact position of the bearing-pulley and the corresponding contact force, respectively. Meanwhile, the change of resistor of the strain gauges can be acquired by the signal sampling circuit of the sensing module. Owing to the SoftBUS framework, all these procedures can be implemented in an autonomous way. The datum of the object's position, the corresponding contact

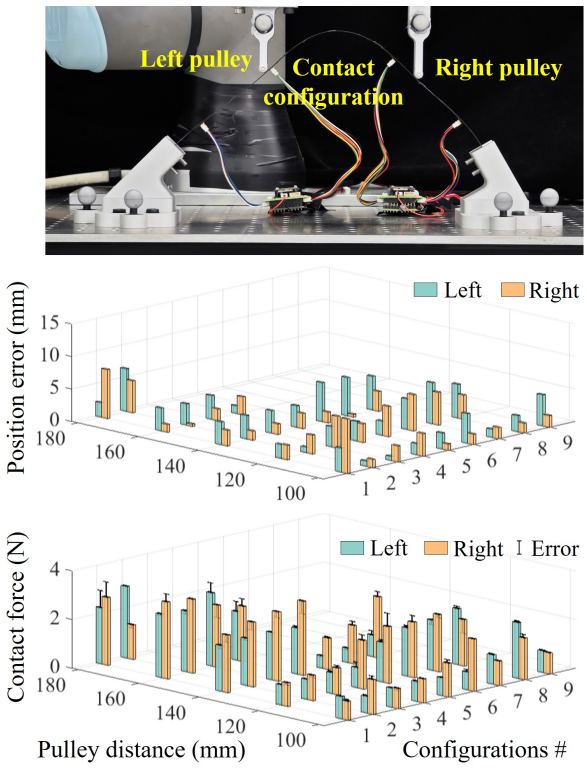

Figure 12. Prediction result of two-point contact.

force, and the local bending of the beam can then be collected simultaneously for identifying the kinetostatic parameters. It should be noted that, in the presented calibration process, the manufacturing errors of the elastic beam and the mounting components are not taken into account. In other words, only the stiffness of the discretized elastic joints, viz. k_i in (3), the line slopes λ_i in (1), and the deploying positions of the strain gauges relating to $\{n_{k,f}, n_{k,l}\}$, are identified exactly according to the measured datum.

For kinetostatic calibration, a 3×19 grid of positions are prescribed for the friction-less pulley, as drawn in Fig. 10, to be in contact with the deformed elastic beam. According to the established kinetostatic model, the joint stiffness, the property of strain gauges and their deploying positions can be corrected in light of the measured datum. First, the relations of the strain gauge's bending angle with respect to the output voltage are provided in Fig. 11. From the results, it can be seen that the linear relationship in (1) is well satisfied. And the proportional coefficients can be specified readily using a linear regression, as illustrated in the figure. It is worth noting that the deploying positions of the strain gauges are adjusted in this step too. On this basis, the structural compliance of the elastic beam can be further refined according to the contact forces measured by the commercial force sensor. Using the identified parameters, the mean estimation error for contact positions is about $1.74\,mm$ in all the measured 57 locations along the deformed beam. Meanwhile, the mean prediction error for contact forces is $0.11\,N$ (about 7.68% relative to the contact force). Hence, the identified parameters are used for contact sensing to improve the prediction accuracy of the developed contact sensor.

5.3 Validation of Two-point Contact Sensing

To validate the effectiveness of the proposed contact sensing methodology, experiments for two-point contact are carried out in this section. As shown in Fig. 12, an adjustable two-pulley end-effector is constructed to apply external contact to the deformed elastic beam. In this case, a set of 30 two-point contact configurations are selected, having the distance of the two pulleys in $\{100, 120, 140, 160, 180\}\,mm$. According to the sensing model derived in Sec. 4.2, the contact positions of the two pulleys can be identified, so are the corresponding contact forces. Meanwhile, the exact contact positions and forces are directly measured by the OptiTrack capture system and the F/T sensor, respectively. A comparison between the contact sensing results and those measured directly can then be made to evaluate the performance on prediction accuracy of the developed contact sensor.

The results of this validation experiment are given in Fig. 12, from which it is known that the two positions of contact can be predicted in a comparatively accurate way, using the developed contact sensor. For the tested 30 configurations of two-point contact, the mean estimation errors for contact positions are $3.38\,mm$ and $2.90\,mm$ for the left and the right pulleys, respectively. Although these errors are slightly larger than those in the scenario of single-point contact presented in Sec. 5.2, they are acceptable comparing to the sensor's overall size. On the other hand, the mean estimation errors for the contact forces are $0.13\,N$ and $0.19\,N$ for the left and right pulleys, respectively, which are about 8.50% and 8.96% with respect to the force magnitudes. From this experiment, both the positions and forces of the two-point contact can be estimated accurately using the developed compact sensor. The perception accuracy and sensing frequency can effectively support the robot for interactive manipulation.

6 Potential Applications in Mobile Robots

Validation experiments have confirmed the accuracy of the proposed methodology. To elucidate its utility within robotic systems, applications for environment perception are provided to demonstrate the potential of the proposed contact sensing method applied in mobile robots. A mobile robot that equipped with the contact sensor is elaborated to exhibit the robot's capability of contact perception. As exhibited in Fig. 13, one piece of the developed contact sensor is deployed around the mobile robot, such that it can perceive the contact with environment via the deformed elastic beam. In this case, the signal of the strain gauges is first acquired on the mobile robot and then wirelessly transmitted to the upper computer via Bluetooth. The contact condition can then be determined in real-time through the sensing algorithm for environment perception and further action. The constructed mobile robot can perform omni-directional motion, and controlled remotely by the upper computer via Bluetooth. It is worth noting that the elastic beam with strain gauges not only serves as a contact sensor, but also acts as a compliant unit for the mobile robot to buffer the external impact.

Here, several scenarios of environment perception and/or task performing are provided to exhibit the effectiveness of the developed contact sensor. In the first scenario, the robot is actuated to make contact with unknown objects and then

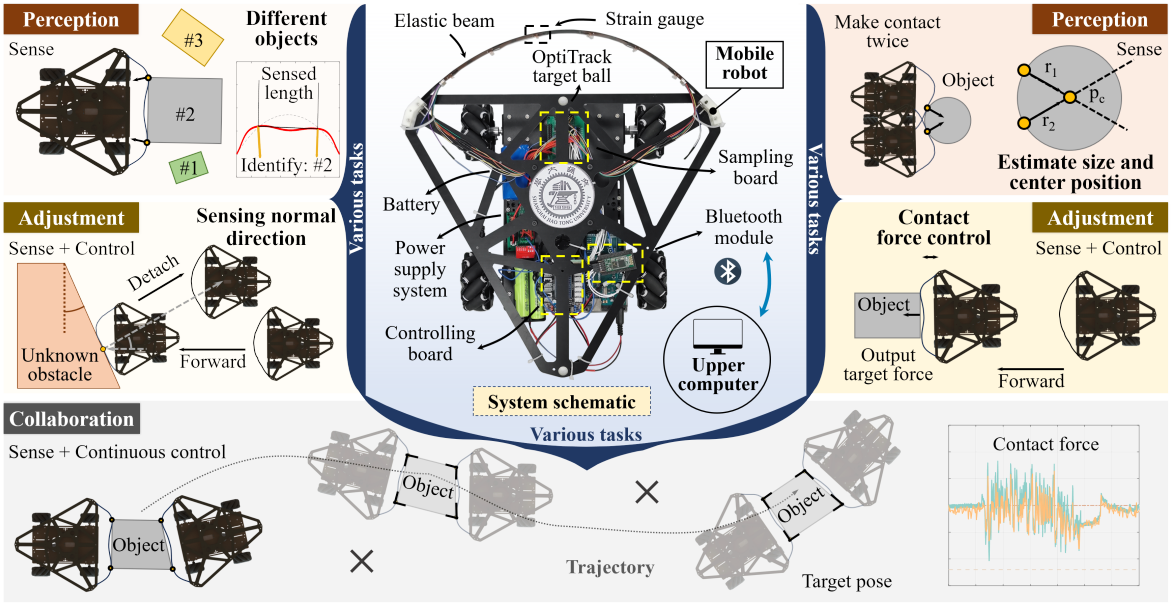

Figure 13. General schematic of environment perception applications of the proposed contact sensing method deployed on mobile robots.

recognize their lengths relying on the capability of contact sensing, as drawn in Fig. 13. Besides, the size and position of a cylindrical object can be identified through two contact trials, allowing the mobile robot to align with the cylinder. Thirdly, the direction of contact is detected using the contact sensor and the robot is driven to move backward to avoid the obstacle. In addition, using a sensor-based control method, the robot is able to buffer the impact by applying controllable force to a fixed object in various conditions. Finally, sensor-based feedback control is exhibited via the collaboration of two mobile robots to move an unstructured object together. In general, a mobile robot equipped with the developed contact sensor is able to recognize objects statically, locate and avoid obstacle dynamically, and manipulate objects continuously.

6.1 Object Length Estimation

As shown in Fig. 14, three boxes of different sizes are used to exhibit the robot's capability of object length estimation. In this case, the robot is driven to make contact with the boxes at the left, middle, and right regions of the deformed beam, respectively. As shown in the figure, two-point contact will occur in this scenario, at the boxes' corners. Thus, the width of the boxes can be estimated in light of the distance of the contact positions. Here, the process of length estimation is repeated for five times for each configuration to validate the effectiveness of the proposed contact sensing method. Hence, there are totally $3 \times 3 \times 5 = 45$ trials performed to identify the size of unknown objects.

The result shows that, in this example, the three boxes of different sizes can be estimated correctly in all the 45 tests, relying on the sensed positions. The estimation accuracy of the objects' lengths is evaluated in a quantitative way, in the form of errors for different boxes at different sensing regions, as drawn in Fig. 14. From the result, the object lengths can be estimated precisely. The overall mean error of size estimation is $2.33\,mm$, which is quite accurate comparing to the box size in the range of $100\,mm \sim 200\,mm$ (within 2%).

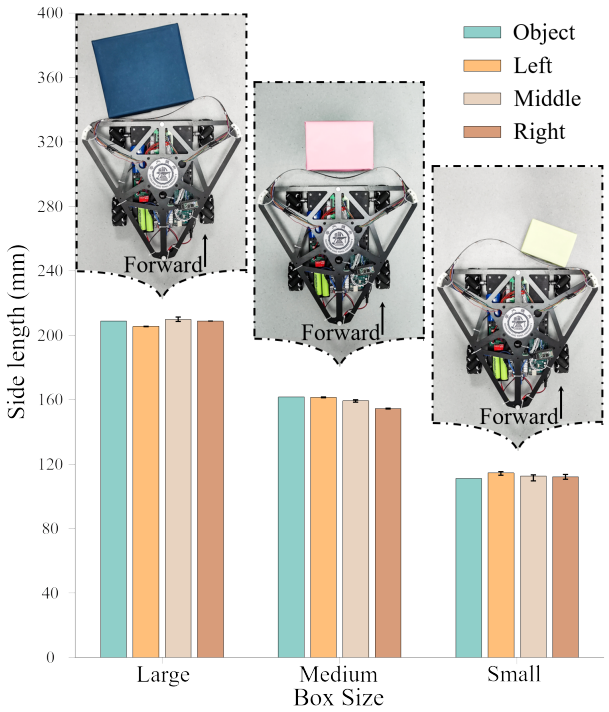

Figure 14. Result of object length estimation.

Moreover, the robot's performance on length estimation is evenly distributed in different regions of the elastic beam and applicable to objects of different sizes. For each tested configuration, the lengths of the tested box is obtained as the average value of five trials, and the corresponding deviations are provided in Fig. 14. From the result, it is known that the performance of the developed contact sensor is stable in the object length estimation across different trials. And the deviations of length estimation are all within $3.70\,mm$ in the five tests for each box at different regions. Meanwhile, the direction of contact can also be identified in such a scenario, which will be used in the following sections.

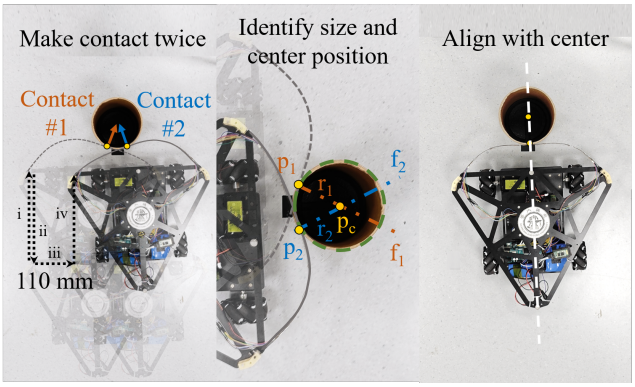

Figure 15. Result of object size identification and localization.

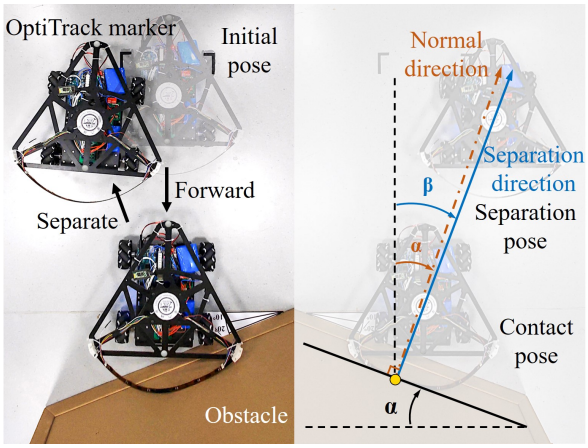

Figure 16. System schematic of obstacle perception.

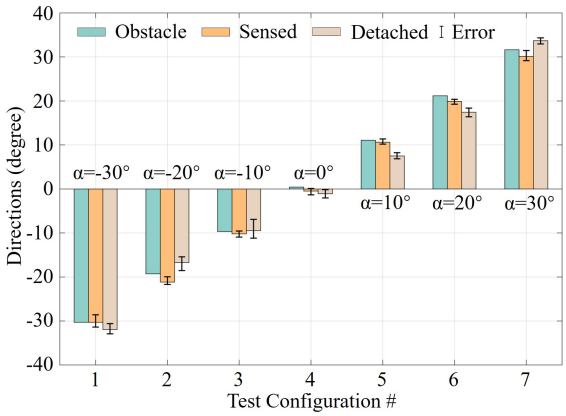

Figure 17. Result of obstacle sensed and separation direction.

6.2 Object Size Identification and Localization

In this scenario, the mobile robot is actuated to make contact with a cylindrical object (of the diameter 130 mm) twice, and then estimate the object's size and locate its position via the developed contact sensor. As shown in Fig. 15, the positions and normal directions of the two contact configurations are acquired to estimate the size and location of the cylindrical object. In this case, the mobile robot moves forward from its initial position to make the first contact for the contact point P_1 and the corresponding normal direction f_1. Thereafter, it moves backward to the initial position, then side-ward for a certain distance (110 mm), and again forward to make the second contact. Analogously, the contact position P_2 and the normal direction f_2 can be perceived via the contact sensor. It should be noted that the encoders of the wheel motors are used to record the trajectory of the mobile robot. Hence, the relative coordinates of the contact locations with respect to the robot's initial position can be obtained for perception of the unknown cylindrical object.

As a result, the center point of the cylindrical object can be determined directly as the intersection of the two normal directions, denoted as P_c. Then, the radius of the object just corresponds to the distance from the contact positions to the center point. Here, the mean value $r = (r_1 + r_2)/2$ is chosen as the estimation of the object's radius, where $r_1 = \overline{P_1 P_c}$ and $r_2 = \overline{P_2 P_c}$. To demonstrate the effectiveness of the proposed method, the mobile robot is then actuated to move directly toward the cylindrical object along the central line.

Five trials are conducted for this example, from different positions with various initial distances. Across these trials, the mean estimation error of the object's radius is 3.66 mm, with a maximum error of 5.61 mm. The mean error for the object position is 10.60 mm, with a maximum of 12.88 mm. Comparing to object size, the estimation the object's radius is quite accurate using the developed contact sensor. Moreover, the mobile robot is running in an open-loop controlling mode for trajectory following, only relying on the feedback of the wheel motors' encoders. As a consequence, its motion errors will influence the sensing accuracy of the object's size and position significantly.

6.3 Obstacle Perception and Separation

Since the developed contact sensor has been integrated on the omni-directional mobile robot, it is able to perceive the contact with the environment and take appropriate actions in real-time. Here, examples of 'contact-perception-separation' are provided to demonstrate the potentials of the developed contact sensor applied in obstacle perception and separation. As shown in Fig. 16, a planar object is placed on the traveling trajectory of the mobile robot to block it as a rigid wall. The contact sensor is then utilized to perceive the occurrence of collision and determine the exact location and direction of the obstacle. Then, the mobile robot can be driven backward to separate from the contact and move towards the direction away from the obstacle.

As illustrated in Fig. 16, the obstacle is set at an angle of α in the traveling direction of the mobile robot. Relying on the deployed contact sensor, the location of contact on the elastic beam can be estimated as a collision occurs. Accordingly, the relative position of the obstacle can be determined for the mobile robot for separation. As indicated in the above, using the proposed contact sensing methodology, the orientation of the planar obstacle can also be identified as the tangent to the deformed beam at the contact point. Hence, the mobile robot can drive backward along the normal direction to separate from the collision. As shown in the figure, a set of locations with different orientation (evenly distributed in $[-30^\circ \sim 30^\circ]$ with an increment 10°) are prescribed for the obstacle. The separation direction, denoted by β, is measured to validate the effectiveness of the proposed contact sensing methodology in collision detection and obstacle separation.

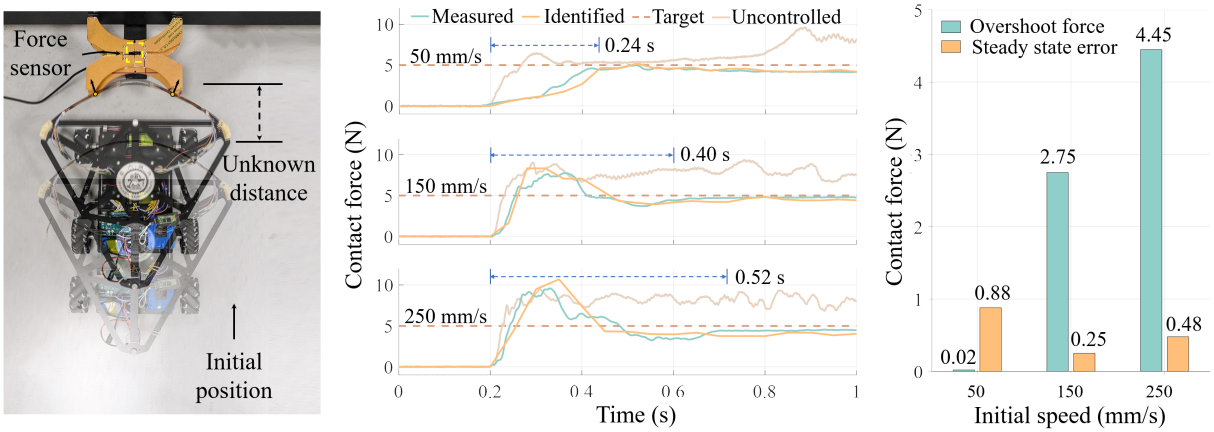

Figure 18. Active contact force control during an unexpected collision.

The mobile robot's performance on collision detection and separation at the tested seven trials, is provided as shown in Fig. 17, in which the obstacle's actual orientations are also provided as a reference. From the results, it can be seen that the mobile robot equipped with the developed contact sensor is able to perceive the contact and move back away from the obstacle accurately. The mean error of the robot's separation motion, the angular deviation of β from α, is 2.41° at the 7 tested locations. It is relatively accurate for a simple omni-directional mobile robot integrating the developed contact sensor as a compact exterior accessory.

Further, the separation accuracy is also influenced by the motion inaccuracy of the mobile robot. Hence, the direction of contact that is directly identified by the contact sensor is also provided in Fig. 17, as a comparison. From the results, it is known that the estimation accuracy of contact direction is much higher than the robot's separating one. Excluding the influence of motion, the mean error can be reduced to 1.19° for obstacle perception, at the tested locations. And this is in accordance with the sensing accuracy given in the validation experiments in Sec. 5.2. As a consequence, with integration of the developed contact sensor, the mobile robot possesses the ability to perceive the obstacle.

6.4 Active Control of Contact Force

Owing to the structural compliance of the elastic beam, the developed contact sensor could endow the mobile robot with the ability to cushion impact during an unexpected collision. Since the contact force can be detected in real-time, a force-based feedback control can be implemented for the robot to cushion the impact in a passive/active hybrid way. As shown in Fig. 18, an unknown object is placed in the travelling path of the mobile robot at an unknown distance. The robot moves forward to make collision with the object at a specific speed. Then, active control of the interactive force takes effect by adjusting the robot's motion according to the sensed force. Here, a PID control theme is adopted to modulate the speed of the mobile robot according to the deviation of the sensed contact force from the target one. In this case, a contact force of $5N$ is prescribed for the target status between the mobile robot and the unknown object via the elastic beam. Different initial speeds ($50mm/s$, $150mm/s$, and $250mm/s$) are tested to validate the effectiveness of the developed contact sensor in collision cushioning. For each speed condition, five trials are conducted to demonstrate the robustness of the proposed contact sensing methodology.

The results are given in Fig. 18, in which the exact contact force during the whole process (measured by a commercial F/T sensor) is provided. From the results, it can be seen that the mobile robot equipped with the developed contact sensor, is able to detect and cushion the impact in collision with the unknown object. The contact forces can be controlled to the target one in all tested conditions of different speeds, with the steady-state errors less than $0.88N$. Besides, the contact forces can be stabilized quickly, with the setting time less than 0.52 seconds. Especially for the low-speed case, it only takes the robot 0.24 seconds to reach a steady state.

It should be noted that the impact of collision can also be cushioned passively by virtue of structural compliance of the elastic beam. The contact forces of direct collisions (without controlling the robot's motion actively) are presented in Fig. 18, as a comparison. From the results, it can be seen that the impact load has been effectively mitigated through the elastic deformation of the contact sensor. This property enables the mobile robot to have sufficient space to reach its actuation limit, which provides favorable conditions for active control. As well, the contact force perceived by the developed sensor are presented in Fig. 18, as a reference. It can be seen that the results obtained from the developed contact sensor are quite close to those measured by the commercial one. Hence, the performance on active control of contact force can be further enhanced by using high-performance mobile robots.

6.5 Force-based Interactive Collaboration

The last scenario of potential applications demonstrated in this section is the interactive collaboration of multiple robots through feedback control based on the sensed contact force. As shown in Fig. 19, a rigid object is handled by two mobile robots through the elastic beams that embedded with strain gauges. In this case, the contact sensors not only provide the exact information of contact status, but also exert sufficient clamping forces to keep the object from falling.

Here, the two robots are controlled in master-slave mode to regulate the fluctuations of clamping forces caused by the motion deviations. The rear vehicle acts as the master robot that moves along a prescribed trajectory, and the front vehicle

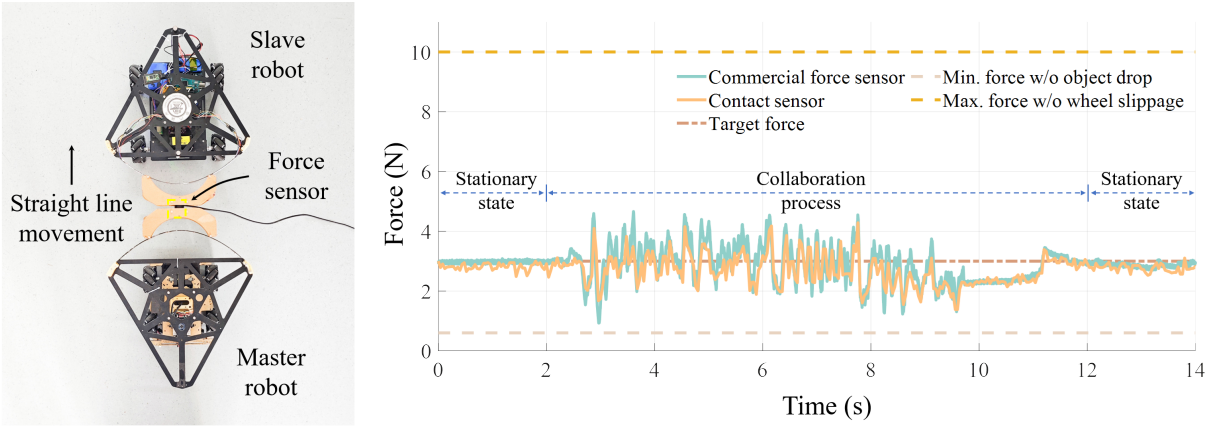

Figure 19. Two-robot collaboration of straight line trajectory.

serves as the slave whose motion rate is actively adjusted in light of the identified interaction force via the contact sensor. It is worth noting that, the movement of the slave robot is a superposition of two motions. On one hand, it moves subject to the prescribed trajectory as same as the master robot does. But in the mean time, its relative position with respect to the master one is adjusted in real-time to keep a stable clamping force. In this case, a $3N$ clamping force is selected for the two collaborative robots, in ensuring the effective operation of the object (larger than $0.6N$), and avoiding the wheels' slippage of the mobile robots (no more than $10N$).

As demonstrated in Fig. 19, a straight-line trajectory is first prescribe for the two robots to carry an object embedded with a commercial force sensor. The collaborative operation is performed in light of the perceived interactive force through the developed contact sensor. Owing to the passive structural compliance and active force control, the two robots succeed in carrying the unknown object along the prescribe trajectory. The exact clamping force exerted on the object during the whole process is measured by the commercial force sensor, as drawn in Fig. 19, in which that obtained via the developed contact sensor is also provided. From the results, it is known that, although the clamping force fluctuates above and below the prescribed target due to motion errors, it can be controlled within a certain range to prevent the object from dropping or the robots from slipping. Moreover, as exhibited in the figure, the clamping force perceived through the developed contact sensor well fits that measured by the commercial sensor. The mean error during the whole collaboration process is $0.67N$, which demonstrates the effectiveness of the proposed contact sensing method in force-based interactive collaboration.

Based on the same collaboration strategy, three trajectories other than the straight line, including a circular arc, a zigzag line, and an S-type curve, as shown in Fig. 20, are elaborated in this experiment to further verify the effectiveness of the proposed sensor-based interactive collaboration. Benefiting from the equipped contact sensor, real-time feedback control can then be implemented for the mobile robots, to stabilize the clamping force, resulting in the successful force-based interaction for multi-robot collaboration.

The trials of force-based interactive collaboration between the two mobile robots are demonstrated in the supplementary video*, in which those without force sensing and regulation are also provided as a comparison. The results indicate that,

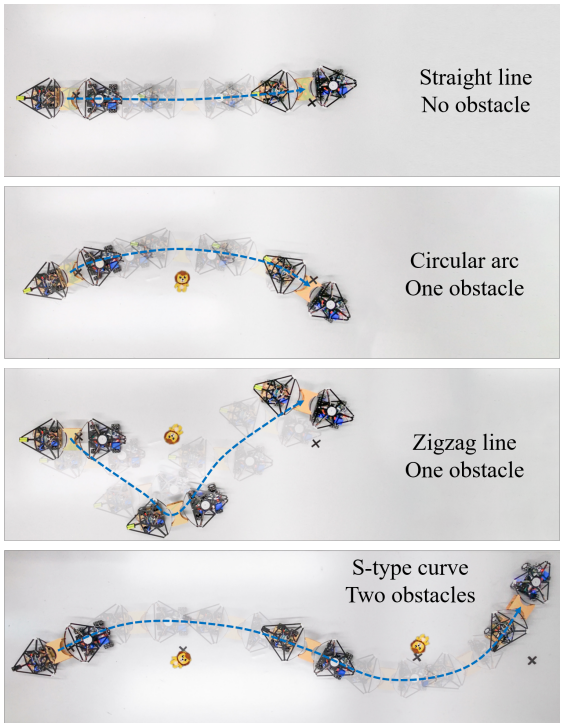

Figure 20. Two-robot collaboration of different trajectories.

relying on the sensor-based feedback control of the clamping force, the object can be successfully carried from its initial position to the target one in all tested trajectories. Otherwise, the collaboration fails during or at the end of the journey, due to the motion inaccuracy caused by speed fluctuations. Thus, the integration of the contact sensor could enable the mobile robots to perceive and interact with each others with the help of contact sensing capability.

6.6 Discussion

As indicated in the above sections, the main contribution of the proposed contact sensing methodology is the integration of passive compliance and sensing capability into the elastic beams. Analytical models are established for the proposed contact sensing methodology, and a compact sampling

*See supplementary materials – S1.

system is constructed for the signal acquiring. It makes the developed sensor efficient in contact sensing and easy for deployment. Hence, a robot equipped with such a contact sensor is able to perform force-based interactive manipulation with the environment and/or other robots. The examples provided in Sec. 6 have demonstrated this feature in an intuitive way, through a mobile robot equipped with the developed contact sensor. The elastic beam integrating strain gauges not only not only acts as a sensor to estimate the contact force, but also as a structure to interact with the environment. The experimental results in Sections 6.4 and 6.5 have demonstrated that the estimated error of the contact force is within an acceptable range, ensuring the success of these tasks. Furthermore, the compliance of the proposed sensor provides not only a smaller force impact, but also more time for the control system to respond. This can also be extended to other interactive operations concerning contact force control.Thus, it will be convenient to promote the developed contact sensor into various robotic systems performing interactive manipulation.

In the future work, the following aspects can be improved for the proposed contact sensing methodology to address the limitation current design. Firstly, only bending deformation of the elastic beam is taken into consideration in the proposed sensing model. As a consequence, the contact forces that are exerted out of the beam plane can not be detected. Model and sensor enhancements can be performed to capture torsional deformations in addition to the bending ones. In addition, compliant parallel mechanisms with spatial motion capacities can be implemented for more complex operations. Secondly, the miniaturization of the utilized strain gauges is limited by the fabrication method for the flexible printed circuit board. Therefore, the maximum number of strain gauges that can be deployed on the elastic beam, is limited by the gauges' size. Furthermore, a hybrid sensing framework combining data-driven approaches with model-based computations could be implemented to significantly enhance system robustness while reducing dependence on strain gauge calibration and material property assumptions. In addition, the elastic beam used for the developed contact sensor is initially straight in the undeformed state. Thus, the distribution of contact stiffness along the whole curved beam, in conjunction with its compliance performance, is uniquely determined according to the geometric constraints at the two ends. Finally, thin carbon-fiber plates are used as the elastic beam in the developed contact sensor, such that the structural stiffness is not very high. And the contact sensor is unable to withstand large interaction force with the environment.

7 Conclusions

In this paper, a simple yet reliable contact sensing method is proposed based on continuum mechanics of slender elastic beams. Unlike most existing methods that directly relying on the pressure-sensitive elements, the developed contact sensor exploits the propagation of contact-induced deformation in a simply supported elastic beam undergoing large deflections. Using a discretization approach, the exact elastostatic model of the constrained elastic beam is established to characterize its force-deformation behavior, as well the mapping between local bending and global deflection. Hence, a small number of strain gauges can endow the beam with the capability of multi-point contact sensing along the whole structure. Owing to the closed-form system Jacobian, large-scale contact can be identified efficiently by solving the nonlinear model using the Newton-Raphson algorithm. Consequently, not only the contact location, but also the magnitude and direction of the reaction force can be determined exactly using the proposed sensing method.

In the developed contact sensor, industry-grade thin-film strain gauges are used to make it compact in structure, and stable for sensing. Meanwhile, a wireless sampling system is constructed for the acquisition and transmission of signal to make the contact sensor easy to integration, and flexible for deployment. The effectiveness of the contact sensing method is validated by a variety of experiments on the perception of multi-point contact. In addition, potential applications of the developed contact sensor in mobile robots are provided to demonstrate the capability of contact sensing, including the recognition of object size, identification of obstacle position, and force-based collaboration. The results indicate that the proposed contact sensing methodology possesses enormous potential in robotic systems for environment perception and interactive manipulation based on contact forces.

Appendix

System Jacobian of the Multi-Contact Model

In the system Jacobian matrix (16), the partial derivative of $\boldsymbol{y}_1$ and $\boldsymbol{y}_2$ with respect to $\boldsymbol{l}$ can be obtained directly as

$$\mathbf{J}_{1,l} = \frac{\partial \boldsymbol{y}_1}{\partial \boldsymbol{l}} = [\tilde{\boldsymbol{u}}_1, \boldsymbol{0}]_{6\times 2}, \ \mathbf{J}_{2,l} = \frac{\partial \boldsymbol{y}_2}{\partial \boldsymbol{l}} = [\boldsymbol{0}, \tilde{\boldsymbol{u}}_2]_{6\times 2}$$

where $\tilde{\boldsymbol{u}}_i = [\boldsymbol{0}^T, \boldsymbol{u}_i^T]$, $i = 1,2$. Here $\boldsymbol{u}_i \in \mathbb{R}^{3\times 1}$ is the unit vector associated with the $\boldsymbol{u}$-axes of $\{\mathrm{C_i}\}$.

Likewise, the one of $\boldsymbol{\tau}$ with respect to the contact location $\boldsymbol{l}$ can be obtained as

$$\mathbf{J}_{\boldsymbol{\tau},l} = \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{l}} = -\sum_{i=1}^{2} f_i \mathbf{J}_i^T \frac{\partial \boldsymbol{W}_i}{\partial \boldsymbol{l}} \Rightarrow \frac{\partial \boldsymbol{W}_i}{\partial \boldsymbol{l}} = \begin{bmatrix} \hat{\mathbf{r}}_i \boldsymbol{\Psi}_i - \hat{\mathbf{v}}_i \boldsymbol{\Gamma}_i \\ \boldsymbol{\Psi}_i \end{bmatrix} \mathbf{J}_{i,l}$$

where $\boldsymbol{\Psi}_i = [-\hat{\mathbf{v}}_i, \mathbf{0}_3] \in \mathbb{R}^{3\times 6}$ and $\boldsymbol{\Gamma}_i = [-\hat{\mathbf{r}}_i, \mathbf{I}_3] \in \mathbb{R}^{3\times 6}$. Here, $\hat{\mathbf{v}}_i$ and $\hat{\mathbf{r}}_i$ represent the 3×3 anti-symmetric matrices of $\boldsymbol{v}_i$ and $\boldsymbol{r}_i$, respectively.

Meanwhile, the partial derivative of $\boldsymbol{\tau}$ with respect to $\boldsymbol{f}$ can be obtained in a similar way as

$$\mathbf{J}_{\boldsymbol{\tau},f} = \frac{\partial \boldsymbol{\tau}}{\partial \boldsymbol{f}} = -\left[\mathbf{J}_1^T \boldsymbol{W}_1, \mathbf{J}_2^T \boldsymbol{W}_2\right]_{n\times 2}$$

In addition, the component relating $\mathbf{J}_i^T \boldsymbol{F}_i$, $(i = 1,2)$ to $\boldsymbol{\theta}$ can be derived as

$$\mathbf{K}_{\mathbf{J}_i} = \frac{\partial \left(\mathbf{J}_i^T \boldsymbol{F}_i\right)}{\partial \boldsymbol{\theta}} = \left[\boldsymbol{F}_i^T \frac{\partial \boldsymbol{\xi}_j}{\partial \boldsymbol{\theta}_k}\right]_{j,k} + \mathbf{J}_i^T \left(\frac{\partial \boldsymbol{F}_i}{\partial \boldsymbol{\theta}}\right) = \mathbf{K}_{\mathbf{J}_{i,J}} + \mathbf{K}_{\mathbf{J}_{i,F}}$$

where the item $\mathbf{K}_{\mathbf{J}_{i,J}}$ can be obtained in the same way as $\mathbf{K}_{\mathbf{J}_t}$ in (6). While the one $\mathbf{K}_{\mathbf{J}_{i,F}}$ can be obtained directly as

$$\mathbf{K}_{\mathbf{J}_{i,F}} = f_i \mathbf{J}_i^T \frac{\partial \boldsymbol{W}_i}{\partial \boldsymbol{\theta}} = f_i \mathbf{J}_i^T \begin{bmatrix} \hat{\mathbf{r}}_i \boldsymbol{\Psi}_i - \hat{\mathbf{v}}_i \boldsymbol{\Gamma}_i \\ \boldsymbol{\Psi}_i \end{bmatrix} \mathbf{J}_i$$

Hence, the system Jacobian matrix of the beam's multi-point contact model can be derived in closed-form.

Funding

This work was jointly supported by the National Key Research & Development Program of China under the Grant 2022YFB4701200, and the Natural Science Foundation of China (NFSC) under the Grants 52022056.

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